

## A Baselines

The first group is graph classification models:

- GCN [8]: a GNN that uses a convolution function on a graph to propagate information within the neighborhood of nodes;
- GraphSAGE [6]: a GNN that leverages a sampling technique to aggregate features from the neighborhood.
- GAT [13]: a GNN that adopts an attention mechanism within the neighborhood of each node;
- GIN [20]: a GNN that follows graph isomorphism to capture the properties of a graph.
- LRGNN [17]: a GNN stacking multiple GNNs to extract the long-range dependencies;
- GRDL [16]: a GNN treating node embeddings as a discrete distribution, enabling direct classification without global pooling.
- BiView [3]: a GNN utilizing both homophily and heterophily for graph classification.

The second group is GAD models:

- iGAD [25]: a GNN with a substructure-aware component to capture properties of anomalous graphs.
- GmapAD [12]: a GNN mapping graphs into a latent space where anomalies can be effectively detected;
- RQGNN [5]: a GNN using Rayleigh Quotient to obtain information from both spectral and spatial spaces.
- UniGAD [10]: a GNN that unifies different levels of graph-related tasks.
- GADTrans [22]: a GNN with transformer-based backbone for graph anomaly detection.

The third group is graph-level augmentation frameworks:

- MAA [23]: a framework using node split and merge, and sub-graph mix to augment graphs heuristically;
- GLA [24]: a framework augmenting data in the representation space from the most difficult direction while keeping the label of augmented data the same as the original samples;
- GM [7]: a framework that interpolates graphons of different classes in the Euclidean space to get mixed graphons;
- FGWM [11]: a framework that seeks a midpoint of source graphs in the Fused Gromov-Wasserstein metric space to interpolate graphons of different classes.
- DP [19]: a framework that uses mask or noise to perturb the spectral spaces of graphs for augmentation.

## B Algorithm

### Algorithm 1: Preprocess

**Input:**  $\mathcal{D}, k_l, k_s$

```

1 for  $G$  in  $\mathcal{D}$  do
2    $G.A \leftarrow \frac{1}{2}(I + G.D^{-\frac{1}{2}} * G.A * G.D^{-\frac{1}{2}});$ 
3    $G.U_l, G.A_l \leftarrow \text{EVD}(G.A, k_l);$ 
4    $G.U_s, G.A_s \leftarrow \text{EVD}(G.A, k_s);$ 

```

## C Time Complexity Analysis

For the Preprocess function, we first analyze the time complexity of the matrix multiplication. Since we utilize sparse matrices to

### Algorithm 2: FGG

**Input:**  $\mathcal{D}, H_l, H_s$

**Output:**  $\mathcal{D}'$

```

1 for  $G$  in  $\mathcal{D}$  do
2   for  $i = 0$  to  $H_l$  do
3      $G_l \leftarrow G_l + \omega_l[i] * G.U_l * G.A_l^{\alpha_l[i]} * G.U_l^T;$ 
4   for  $i = 0$  to  $H_l$  do
5      $G_s \leftarrow G_s + \omega_s[i] * G.U_s * G.A_s^{\alpha_s[i]} * G.U_s^T;$ 
6    $G' \leftarrow \omega * G_l + (1 - \omega) * G_s;$ 
7    $\mathcal{D}' \leftarrow \mathcal{D}' \cup G';$ 
8 Return  $\mathcal{D}';$ 

```

### Algorithm 3: WDML

**Input:**  $f, \mathcal{D}, \mathcal{D}'$

```

1 for  $G, G'$  in  $\mathcal{D}, \mathcal{D}'$  do
2    $s, o \leftarrow f(G);$ 
3    $s', o' \leftarrow f(G');$ 
4    $m \leftarrow \frac{1 - \cos(o, o')}{2};$ 
5    $L_{\text{WDML}} \leftarrow L_{\text{WDML}} + \frac{1}{N_{G.y}} \log \frac{e^{s[G.y] - m}}{e^{s[G.y] - m} + e^{s[1 - G.y]}};$ 
6  $L_{\text{WDML}}.\text{backward}();$ 

```

### Algorithm 4: MVP

**Input:**  $f, \mathcal{D}, \mathcal{D}', \tau_n, \tau_a$

**Output:**  $\mathcal{D}''$

```

1 for  $G, G'$  in  $\mathcal{D}, \mathcal{D}'$  do
2    $s, o \leftarrow f(G);$ 
3    $s', o' \leftarrow f(G');$ 
4   if  $s[0] < \tau_n \wedge s'[0] \leq \tau_n$  then
5      $G.y \leftarrow 0;$ 
6      $\mathcal{D}'' \leftarrow \mathcal{D}'' \cup G;$ 
7   else if  $s[1] < \tau_a \wedge s'[1] \leq \tau_a$  then
8      $G.y \leftarrow 1;$ 
9      $\mathcal{D}'' \leftarrow \mathcal{D}'' \cup G;$ 
10 Return  $\mathcal{D}'';$ 

```

### Algorithm 5: FracAug

**Input:**  $f, \mathcal{D}_{\text{train}}, \mathcal{D}_{\text{val}}, \mathcal{D}_{\text{test}}, H_l, H_s, k_l, k_s, e_{\text{warmup}}, e_{\text{aug}}, \tau_n, \tau_a$

```

1  $\text{Preprocess}(\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, k_l, k_s);$ 
2  $\mathcal{D}'_{\text{train}} \leftarrow \mathcal{D}_{\text{train}};$ 
3 for  $e = 0$  to  $e_f$  do
4   if  $e > e_{\text{warmup}} \wedge e \% e_{\text{aug}} == 0$  then
5     for  $e' = 0$  to  $e_{\text{FGG}}$  do
6        $\mathcal{D}_{\text{temp}} \leftarrow \text{FGG}(\mathcal{D}_{\text{train}}, H_l, H_s);$ 
7        $\text{WDML}(f, \mathcal{D}, \mathcal{D}_{\text{temp}});$ 
8        $\mathcal{D}_{\text{temp}} \leftarrow \text{FGG}(\mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, H_l, H_s);$ 
9        $\mathcal{D}_{\text{temp}} \leftarrow \text{MVP}(f, \mathcal{D}_{\text{val}} \cup \mathcal{D}_{\text{test}}, \mathcal{D}_{\text{temp}}, \tau_n, \tau_a);$ 
10       $\mathcal{D}'_{\text{train}} \leftarrow \mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{temp}};$ 
11  $\text{train}(f, \mathcal{D}'_{\text{train}});$ 

```

**Table 9: Comparison of average running time (s).**

| Datasets | MAAv+FA       | NS      | SM     | GLAv+FA       | GLA     | GMv+FA        | GM    | FGWMv+FA      | FGWM   |
|----------|---------------|---------|--------|---------------|---------|---------------|-------|---------------|--------|
| MCF-7    | 92.18+58.56   | 1282.84 | 876.39 | 92.00+73.57   | 1493.55 | 90.54+107.51  | 51.64 | 92.29+106.76  | 553.90 |
| MOLT-4   | 133.10+83.70  | 1352.07 | 881.00 | 128.18+113.70 | 1924.35 | 131.30+150.27 | 64.55 | 129.84+176.65 | 855.30 |
| PC-3     | 91.26+70.05   | 1249.30 | 879.76 | 90.55+64.84   | 1484.25 | 90.46+110.26  | 51.52 | 90.70+98.88   | 613.19 |
| SW-620   | 133.80+102.96 | 1248.39 | 875.52 | 136.75+99.17  | 2076.36 | 133.42+158.47 | 75.68 | 133.50+153.30 | 873.41 |
| NCI-H23  | 130.20+108.73 | 1301.09 | 883.09 | 130.63+93.86  | 2091.18 | 131.54+134.20 | 63.68 | 131.07+143.53 | 896.41 |
| OVCAR-8  | 131.48+98.46  | 1351.17 | 890.68 | 132.62+102.36 | 2077.41 | 132.07+168.57 | 61.41 | 131.23+130.47 | 986.01 |

**Table 10: Comparison of average wall-clock time (s).**

| Datasets      | MAAv+FA        | NS      | SM      | GLAv+FA       | GLA     | GMv+FA        | GM     | FGWMv+FA      | FGWM    |
|---------------|----------------|---------|---------|---------------|---------|---------------|--------|---------------|---------|
| MCF-7         | 135.73+69.63   | 1410.86 | 912.23  | 136.19+92.94  | 1543.64 | 131.46+107.54 | 70.60  | 132.19+106.70 | 949.81  |
| MOLT-4        | 191.15+93.22   | 1433.10 | 936.19  | 194.52+127.01 | 1993.29 | 189.00+155.83 | 98.79  | 188.37+164.47 | 1487.41 |
| PC-3          | 133.72+74.67   | 1423.05 | 973.39  | 132.97+86.91  | 1524.84 | 131.72+116.66 | 68.41  | 131.70+103.64 | 968.09  |
| SW-620        | 193.47+104.12  | 1411.45 | 945.87  | 191.01+130.31 | 2138.65 | 194.62+162.24 | 102.90 | 191.97+162.04 | 1321.40 |
| NCI-H23       | 191.622+113.72 | 1417.68 | 896.13  | 191.50+118.83 | 2220.64 | 192.70+139.89 | 102.48 | 194.30+150.65 | 1438.07 |
| OVCAR-8       | 194.69+102.93  | 1459.6  | 891.68  | 192.11+127.31 | 2521.36 | 190.40+160.44 | 102.89 | 193.93+171.23 | 1430.77 |
| P388          | 171.61+103.73  | 1573.25 | 933.75  | 172.97+130.55 | 2146.63 | 171.64+146.75 | 104.16 | 175.47+195.82 | 1754.24 |
| SF-295        | 193.68+93.42   | 1640.00 | 889.18  | 195.20+129.01 | 2065.14 | 190.63+165.59 | 102.69 | 194.14+185.80 | 1469.35 |
| SN12C         | 191.82+102.57  | 1429.26 | 972.91  | 192.14+125.52 | 2116.96 | 192.33+145.00 | 101.46 | 192.01+185.39 | 1378.80 |
| UACC257       | 191.56+100.38  | 1429.63 | 977.20  | 193.70+122.99 | 2110.75 | 192.43+159.39 | 102.32 | 189.07+164.62 | 1320.10 |
| PROTEINS_full | 12.90+7.600    | 213.95  | 216.55  | 12.28+8.87    | 80.33   | 12.91+9.25    | 6.65   | 12.04+8.55    | 42.86   |
| DBLP_v1       | 69.62+67.53    | 2238.69 | 1866.97 | 66.97+74.43   | 3155.02 | 69.79+99.96   | 61.35  | 66.85+104.41  | 227.15  |

conduct the experiment, the time complexity of the multiplication is  $O(\text{nnz}(G.D^{-\frac{1}{2}}) * \text{nnz}(G.A) + \text{nnz}(G.D^{-\frac{1}{2}} * G.A) * \text{nnz}(G.D^{-\frac{1}{2}}))$ , where  $\text{nnz}$  means non-zero entries of the matrix. Then, by adopting EVD to only keep the top- $k_l$  largest and top- $k_s$  smallest eigenvalues, the time complexity can be  $O(n * (k_l^2 + k_s^2) + m * (k_l + k_s))$ , where  $n, m$  is the number of nodes/edges. As shown in Algorithm 5, Preprocess can be called before the training process, and thus it won't burden the training or inference of our FracAug.

Then, we analyze the time complexity of FGG for each graph. As presented in Algorithm 2, we perform sparse matrix multiplication for every sample  $H_l$  and  $H_s$  times. Besides, since  $G.A$  is a diagonal matrix, the time complexity of multiplying  $G.A$  is the same as that of multiplying  $G.A^a$ . Therefore, the total time complexity of FGG is  $O(\text{FGG}) = O(\text{nnz}(G.U_l) * k_l + \text{nnz}(G.U_l * G.A_l) * \text{nnz}(G.U_l^T) + \text{nnz}(G.U_s) * k_s + \text{nnz}(G.U_s * G.A_s) * \text{nnz}(G.U_s^T))$ .

Next, we analyze the time complexity of WDML in Algorithm 3. Assuming that we only have one sample in  $\mathcal{D}_{train}$ , then the time complexity of WDML is  $O(\text{WDML}) = O(d)$ , where  $d$  is the dimension of the generated graph embedding  $\mathbf{o}$ .

Moreover, as shown in Algorithm 4, in MVP, we only need to see if the probability predicted by the given GNN satisfies the criterion, so the time complexity for MVP is  $O(\text{MVP}) = O(1)$ .

Finally, in Algorithm 5, we combine all the time complexities together within one training epoch of the given GNN  $f$ , assuming the time complexity of  $f$  for each sample is  $O(f)$ , then we have the total complexity as  $O(e_{\text{FGG}} * (O(\text{FGG}) + O(\text{WDML}) + O(f)) * N_{train} + (O(\text{FGG}) + O(\text{MVP}) + O(f)) * (N_{val} + N_{test}))$ , where  $N_{val}, N_{test}$  represent the number of samples in  $\mathcal{D}_{val}$  and  $\mathcal{D}_{test}$ , respectively.

In practice, we set  $e_{\text{FGG}}$  to 10 and  $e_{\text{aug}}$  to 25, which can reduce the computational cost, and FGG can still converge. According to the final complexity, we can see the dominant factor within each epoch is  $O(e_{\text{FGG}} * O(f) * N_{train} + O(f) * (N_{val} + N_{test}))$ . For such

a factor, we need to calculate it in total  $\frac{ef - e_{\text{warmup}}}{e_{\text{aug}}} * e_{\text{FGG}}$  times, which is much less than the original training epoch of  $f$ . Hence, the increase in time complexity will not be the limitation of our FracAug in real applications.

In Tables 9 and 10, we present a detailed runtime comparison and wall-clock time comparison between FracAug and several leading graph augmentation methods, respectively. For each technique, we decompose the total computational cost into a one-time preprocessing phase, performed once per dataset, and the subsequent training time measured over multiple epochs. While some baselines require repeated feature perturbations or costly online sampling at every iteration, FracAug's eigenvalue decomposition is only performed during preprocessing. As a result, the per-epoch training overhead of FracAug remains on par with, or even below, that of competing approaches, despite leveraging additional spectral information to boost anomaly detection performance.

Crucially, these efficiency gains do not come at the expense of detection performance. Across all datasets and baseline comparisons, FracAug consistently delivers state-of-the-art AUROC, AUPRC, and F1-score results while maintaining competitive total runtimes. By amortizing the heavier spectral computations over the entire training cycle and by implementing optimized matrix operations, FracAug strikes an effective balance between computational tractability and augmentation quality. This combination of speed and performance underscores the practical value of our method: practitioners can readily adopt FracAug for real graph applications without incurring prohibitive time costs.

## D Memory Complexity Analysis

In Tables 11 and 12, we report detailed comparisons of total and peak memory consumption for FracAug against several leading graph augmentation methods. For each method in Table 7, we further

**Table 11: Comparison of average consumed memory (MB).**

| Datasets | MAAv+FA       | NS     | SM     | GLAv+FA       | GLA     | GMv+FA        | GM     | FGWMv+FA      | FGWM   |
|----------|---------------|--------|--------|---------------|---------|---------------|--------|---------------|--------|
| MCF-7    | 171.99+530.47 | 655.12 | 689.79 | 171.88+687.56 | 1125.73 | 172.20+538.38 | 675.25 | 172.32+552.62 | 598.73 |
| MOLT-4   | 240.71+534.24 | 763.71 | 784.25 | 239.91+672.44 | 1262.11 | 239.92+538.74 | 770.23 | 239.85+573.34 | 817.38 |
| PC-3     | 170.76+530.65 | 652.03 | 687.16 | 170.68+661.78 | 1118.31 | 170.50+533.72 | 671.04 | 170.77+505.36 | 602.24 |
| SW-620   | 243.48+530.75 | 764.33 | 801.05 | 241.11+666.00 | 1276.12 | 243.46+531.57 | 769.41 | 249.02+546.39 | 838.00 |
| NCI-H23  | 248.02+531.50 | 761.09 | 798.24 | 247.96+678.73 | 1300.57 | 250.67+534.30 | 772.79 | 247.66+549.04 | 836.02 |
| OVCAR-8  | 248.82+526.89 | 764.66 | 800.23 | 248.93+669.54 | 1247.54 | 243.27+536.46 | 771.64 | 243.43+550.80 | 805.35 |

**Table 12: Comparison of average peak memory (MB).**

| Datasets      | MAAv+FA | NS    | SM    | GLAv+FA | GLA    | GMv+FA | GM    | FGWMv+FA | FGWM   |
|---------------|---------|-------|-------|---------|--------|--------|-------|----------|--------|
| MCF-7         | 62.53   | 55.84 | 52.07 | 73.49   | 143.10 | 60.55  | 51.83 | 65.46    | 137.32 |
| MOLT-4        | 89.45   | 78.66 | 73.17 | 95.55   | 192.97 | 86.34  | 79.51 | 90.78    | 194.61 |
| PC-3          | 61.93   | 54.51 | 51.37 | 71.91   | 142.55 | 60.67  | 51.40 | 72.49    | 137.59 |
| SW-620        | 91.19   | 80.08 | 74.80 | 85.14   | 196.15 | 93.25  | 81.22 | 93.69    | 197.01 |
| NCI-H23       | 90.79   | 79.15 | 74.23 | 99.11   | 195.42 | 87.77  | 81.13 | 96.70    | 197.96 |
| OVCAR-8       | 91.16   | 80.31 | 74.50 | 97.86   | 196.09 | 92.59  | 81.50 | 94.43    | 195.91 |
| P388          | 93.28   | 82.13 | 76.83 | 103.88  | 200.72 | 95.81  | 90.00 | 96.29    | 208.33 |
| SF-295        | 90.61   | 78.46 | 74.34 | 94.73   | 195.08 | 86.58  | 81.05 | 95.56    | 196.82 |
| SN12C         | 90.02   | 78.43 | 73.81 | 96.41   | 193.95 | 90.44  | 80.36 | 94.84    | 199.04 |
| UACC257       | 89.98   | 79.26 | 74.08 | 96.52   | 193.88 | 88.65  | 80.81 | 90.62    | 194.10 |
| PROTEINS_full | 4.89    | 5.83  | 5.63  | 4.97    | 6.51   | 5.01   | 7.02  | 4.95     | 7.35   |
| DBLP_v1       | 45.35   | 38.40 | 37.67 | 48.54   | 95.30  | 43.37  | 29.05 | 46.59    | 93.41  |

**Table 13: Hyperparameters of 12 datasets based on GIN.**

| Datasets     | MCF-7 | MOLT-4 | PC-3 | SW-620 | NCI-H23 | OVCAR-8 | P388 | SF-295 | SN12C | UACC257 | PROTEINS_full | DBLP_v1 |
|--------------|-------|--------|------|--------|---------|---------|------|--------|-------|---------|---------------|---------|
| $k_l$        | 4     | 4      | 4    | 4      | 3       | 3       | 4    | 3      | 4     | 4       | 3             | 4       |
| $H_l$        | 4     | 3      | 3    | 3      | 3       | 3       | 4    | 4      | 3     | 4       | 4             | 4       |
| $k_s$        | 3     | 4      | 3    | 3      | 4       | 3       | 4    | 3      | 3     | 4       | 3             | 3       |
| $H_s$        | 4     | 4      | 3    | 3      | 3       | 3       | 4    | 3      | 3     | 4       | 4             | 3       |
| $e_{warmup}$ | 50    | 25     | 50   | 50     | 25      | 25      | 50   | 50     | 25    | 50      | 50            | 25      |

break down the total memory usage into a one-time preprocessing stage, executed once per dataset, and the recurring memory cost measured across training epochs. For each method in Table 12, we present the peak memory during the training procedure. Unlike baselines that repeatedly perturb features or perform expensive online sampling at every iteration, FracAug requires eigenvalue decomposition only during preprocessing. Consequently, its per-epoch training overhead remains comparable to, or even lower than, that of competing methods, despite incorporating additional spectral information to enhance anomaly detection.

Importantly, these memory efficiency benefits do not compromise detection quality. Across all datasets and baselines, FracAug consistently achieves state-of-the-art AUROC, AUPRC, and F1-scores while maintaining competitive end-to-end memory cost. By amortizing the heavier spectral computations across the full training process and employing optimized matrix operations, FracAug achieves a strong balance between memory efficiency and augmentation quality. This blend of memory usage and performance highlights the method’s practical value: practitioners can deploy FracAug in real-world graph applications without incurring prohibitive memory costs.

## E Experimental Settings

Table 13 provides a comprehensive summary of the hyperparameter configurations used in our experiments based on the GIN backbone. To obtain the final settings for FracAug, we perform a grid search and select the configuration that achieves the highest combined AUROC, AUPRC, and F1-score on the validation set, after which we report the corresponding test performance. Concretely, the parameters  $k_l, H_l, k_s, H_s$  are each chosen from the set 3, 4, and the number of GNN warmup epochs is selected from 25, 50, which helps keep the overall search cost manageable while still ensuring sufficient coverage of plausible configurations. For fairness and reproducibility, all experiments are conducted on an NVIDIA Quadro RTX 8000, ensuring a consistent computational environment across all evaluations.

## F Margin Loss Comparison

Next, we examine the performance of FracAug under different margin losses, as discussed in Section 4.3, by conducting comparisons across 12 datasets using the GIN backbone. As reported in Table 14, FracAug consistently surpasses its variants that employ Softmax, LMCL, or LDAM as alternative margin formulations. This consistent advantage highlights the effectiveness of our design and

**Table 14: Comparison of different margin loss.**

| Datasets | Metrics | GIN    | +FA    | Softmax | LMCL   | LDAM   |
|----------|---------|--------|--------|---------|--------|--------|
| MCF-7    | AUC     | 0.5867 | 0.5976 | 0.5844  | 0.5880 | 0.5844 |
|          | AUPRC   | 0.2830 | 0.2971 | 0.2781  | 0.2847 | 0.2796 |
|          | MF1     | 0.5366 | 0.5421 | 0.5378  | 0.5372 | 0.5360 |
| MOLT-4   | AUC     | 0.5733 | 0.5854 | 0.5760  | 0.5797 | 0.5751 |
|          | AUPRC   | 0.2830 | 0.3001 | 0.2969  | 0.2952 | 0.2857 |
|          | MF1     | 0.5072 | 0.5103 | 0.4991  | 0.5059 | 0.5076 |
| PC-3     | AUC     | 0.5969 | 0.6119 | 0.6021  | 0.6037 | 0.6001 |
|          | AUPRC   | 0.2797 | 0.2893 | 0.2809  | 0.2778 | 0.2769 |
|          | MF1     | 0.5063 | 0.5205 | 0.5134  | 0.5195 | 0.5144 |
| SW-620   | AUC     | 0.5938 | 0.6004 | 0.5947  | 0.5936 | 0.5931 |
|          | AUPRC   | 0.2776 | 0.2813 | 0.2779  | 0.2768 | 0.2720 |
|          | MF1     | 0.5090 | 0.5155 | 0.5100  | 0.5092 | 0.5133 |
| NCI-H23  | AUC     | 0.5897 | 0.5968 | 0.5913  | 0.5896 | 0.5887 |
|          | AUPRC   | 0.2566 | 0.2659 | 0.2650  | 0.2612 | 0.2622 |
|          | MF1     | 0.5059 | 0.5073 | 0.5001  | 0.5012 | 0.4990 |
| OVCAR-8  | AUC     | 0.5935 | 0.5963 | 0.5905  | 0.5890 | 0.5932 |
|          | AUPRC   | 0.2573 | 0.2612 | 0.2557  | 0.2570 | 0.2579 |
|          | MF1     | 0.5118 | 0.5123 | 0.5087  | 0.5051 | 0.5107 |

reinforces the idea that sample-specific decision boundaries provide a more suitable and flexible mechanism than fixed margins for graph-level anomaly detection. The results align closely with our theoretical analysis in Section 4.3, further validating the motivation and significance of adopting adaptive, distance-aware margins within the FracAug framework.

## G Performance with More Training Data

To further demonstrate the superior capability of our proposed FracAug framework, we conduct additional experiments on PROTEINS\_full and DBLP\_v1 under varying training sizes (%), as illustrated in Figures 5 and 6. Across all training-size configurations, the red curves, representing the performance of GIN equipped with FracAug, consistently appear at the top of the plots. This clear and repeated trend shows that as more training data becomes available, FracAug continues to deliver steady performance gains over the baseline in terms of AUROC, AUPRC, and F1-score. Overall, these results provide additional evidence that FracAug offers a robust, broadly applicable, and practical enhancement to existing models, maintaining its advantages even as the amount of supervised data varies.

In addition, we examine the inductive setting, where only the validation set is used for pseudo-labeling, to evaluate the generalization capabilities of our proposed framework, FracAug. This analysis is represented by the yellow curves in Figures 5 and 6. Across all training-size configurations, the yellow curves, which depict the performance of the GIN model enhanced with FracAug under this setting, consistently outperform the vanilla model. While the yellow curve falls slightly below the red curve due to the reduced amount of data available, the performance gap is minimal. This further underscores the viability of our proposed framework in strictly inductive scenarios. Overall, these findings highlight that FracAug provides a robust, versatile, and practical improvement to existing models, maintaining its effectiveness even when limited to the validation set for pseudo-labeling.

## H Learned Parameters

Here, we present the learned parameters based on GIN,  $\alpha_l$ ,  $\omega_l$ ,  $\alpha_s$ ,  $\omega_s$ , and  $\omega$ , as summarized in Table 15. Specifically,  $\alpha_l$  and  $\alpha_s$  correspond to the learned fractional powers associated with the largest and smallest eigenvalue groups, respectively, while  $\omega_l$  and  $\omega_s$  denote the learned coefficients applied to the fractional graphs generated from these eigenvalues. The vector  $\omega$  represents the balancing coefficient between the two groups of fractional graphs derived from the largest and smallest eigenvalues.

As described in Section 4.2, the hyperparameters  $H_l$  and  $H_s$  determine the number of fractional graphs combined within each eigenvalue group. Consequently, the dimensionality of  $\alpha_l$  and  $\omega_l$  equals  $H_l$ , and that of  $\alpha_s$  and  $\omega_s$  equals  $H_s$ . Since we only consider two eigenvalue groups, the largest and the smallest, the vector  $\omega$  always contains two entries. Moreover, as indicated by Table 13, the optimal values of  $H_l$  and  $H_s$  vary across datasets, leading to corresponding variations in the number of entries reported in Table 15. In cases where a particular entry does not exist due to the selection of  $H_l$  or  $H_s$ , we mark it with “–” for clarity.

From Table 15, we observe that the learned fractional powers fall within the range (0, 3), which is consistent with common practices in GNN design: excessively large effective depths can cause over-smoothing, whereas overly small depths may lead to under-fitting. This observation further validates the rationality of our proposed FracAug, demonstrating that the learned parameters remain well-aligned with established principles while effectively capturing dataset-specific structural information.

## I Additional Experimental Results

We present the additional experimental results for MCF-7, MOLT-4, PC-3, SW-620, NCI-H23, and OVCAR-8 to further demonstrate our superiority over other baselines. On one hand, as shown in Tables 16 and 17, FracAug consistently boosts the performance of a variety of backbone models in both the graph classification and graph-level anomaly detection methods. This pattern holds across all the newly included datasets, reinforcing the generalization ability of our proposed framework and showing that its benefits are not restricted to specific architectures or data distributions. On the other hand, as shown in Table 18, FracAug also outperforms other existing graph augmentation techniques, even when those techniques are evaluated alongside their own native backbones. This result highlights the effectiveness and robustness of our framework, indicating that FracAug can provide complementary advantages regardless of the underlying model design or augmentation strategy used.

## J Standard Deviation for Experimental Results

Due to the limited space, we present the standard deviation for our experimental results here in Tables 19, 20, and 21. Across the three tables reporting standard deviations, for graph classification, graph-level anomaly detection, and data augmentation, we observe a consistent stabilizing effect introduced by FracAug. Although FracAug serves as a plug-in framework that can be seamlessly integrated with diverse baselines, its impact is both clear and substantial. In most cases, the incorporation of FracAug not only improves AUROC, AUPRC, and F1-score, as shown in Tables 16, 17, and 18, but also reduces the variance of these metrics, as shown in Tables

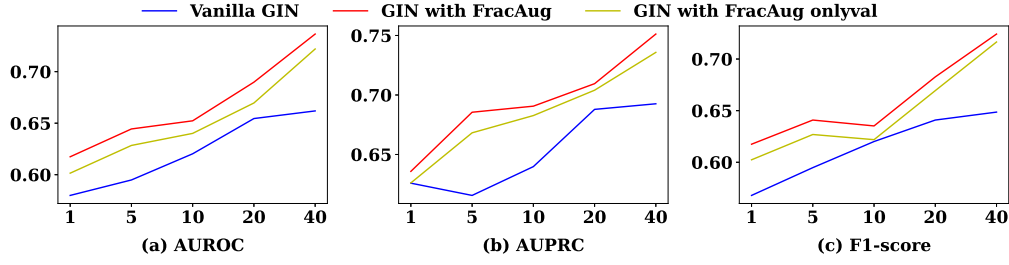


Figure 5: Varying training size (%) for PROTEINS\_full.

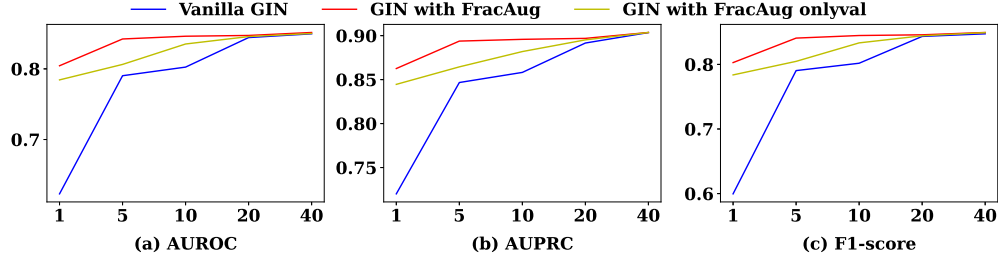


Figure 6: Varying training size (%) for DBLP\_v1.

Table 15: Learned parameters

| Datasets   | MCF-7  | MOLT-4 | PC-3   | SW-620 | NCI-H23 | OVCAR-8 | P388   | SF-295 | SN12C  | UACC257 | PROTEINS_full | DBLP_v1 |
|------------|--------|--------|--------|--------|---------|---------|--------|--------|--------|---------|---------------|---------|
| $\alpha_l$ | 0.9062 | 1.4187 | 1.0862 | 2.186  | 0.4991  | 0.8986  | 2.8806 | 0.8845 | 2.7404 | 1.6178  | 1.1407        | 1.8663  |
|            | 2.0063 | 2.2813 | 1.9282 | 1.5108 | 0.3535  | 0.1262  | 2.9620 | 2.9038 | 0.5877 | 1.9339  | 0.6885        | 1.1374  |
|            | 1.4770 | 1.7911 | 1.9928 | 1.8073 | 2.3672  | 1.1480  | 2.3129 | 2.3939 | 1.0219 | 2.4874  | 2.5432        | 1.1843  |
|            | 1.4269 | -      | -      | -      | -       | -       | 1.2674 | 2.4786 | -      | 2.2634  | 1.4364        | 2.8254  |
| $\omega_l$ | 0.2921 | 0.2655 | 0.3450 | 0.2667 | 0.2793  | 0.3476  | 0.1639 | 0.3051 | 0.2385 | 0.1165  | 0.3842        | 0.1685  |
|            | 0.3108 | 0.3101 | 0.3349 | 0.5068 | 0.4612  | 0.3615  | 0.3654 | 0.1747 | 0.3833 | 0.2806  | 0.1535        | 0.1714  |
|            | 0.1549 | 0.4244 | 0.3202 | 0.2266 | 0.2595  | 0.2909  | 0.2728 | 0.2145 | 0.3782 | 0.3028  | 0.2927        | 0.4004  |
|            | 0.2422 | -      | -      | -      | -       | -       | 0.1979 | 0.3057 | -      | 0.3000  | 0.1696        | 0.2597  |
| $\alpha_s$ | 2.3128 | 1.9621 | 2.8048 | 1.5579 | 2.4828  | 0.5030  | 1.5344 | 1.6912 | 2.9331 | 0.8959  | 1.1290        | 1.7201  |
|            | 2.1539 | 0.4087 | 1.1722 | 1.8088 | 2.1366  | 1.4235  | 1.552  | 1.9896 | 1.5150 | 1.9589  | 0.1467        | 2.8101  |
|            | 0.7485 | 2.9168 | 1.6098 | 0.4474 | 2.2668  | 2.6830  | 2.0318 | 1.8741 | 2.9352 | 2.8844  | 0.5492        | 2.0935  |
|            | 2.5362 | 0.2449 | -      | -      | -       | -       | 2.7135 | -      | -      | 0.1148  | 0.0124        | -       |
| $\omega_s$ | 0.3042 | 0.2821 | 0.3910 | 0.3756 | 0.3292  | 0.3384  | 0.1696 | 0.4052 | 0.4554 | 0.2916  | 0.2407        | 0.2517  |
|            | 0.2569 | 0.2112 | 0.2750 | 0.4245 | 0.3630  | 0.3436  | 0.4001 | 0.2386 | 0.2343 | 0.2539  | 0.3285        | 0.4758  |
|            | 0.2764 | 0.1519 | 0.3340 | 0.1999 | 0.3078  | 0.3180  | 0.2027 | 0.3562 | 0.3103 | 0.3122  | 0.2524        | 0.2725  |
|            | 0.1625 | 0.3548 | -      | -      | -       | -       | 0.2277 | -      | -      | 0.1423  | 0.1785        | -       |
| $\omega$   | 0.4634 | 0.6795 | 0.6481 | 0.5579 | 0.5263  | 0.4541  | 0.4408 | 0.5589 | 0.5724 | 0.5309  | 0.7134        | 0.5174  |
|            | 0.5366 | 0.3205 | 0.3519 | 0.4421 | 0.4737  | 0.5459  | 0.5592 | 0.4411 | 0.4276 | 0.4691  | 0.2866        | 0.4826  |

19, 20, and 21. This dual benefit indicates that FracAug enhances both the effectiveness and the reliability of downstream models. By pseudo-labeling the semantic-preserving graphs, FracAug helps models converge to more stable decision boundaries. Consequently, the tables collectively demonstrate that FracAug is not just a performance booster but also a variance reducer, offering practitioners a method that delivers stronger and more consistent results across a wide range of graph-level anomaly detection tasks.

## K Additional Ablation Study

Beyond the 6 datasets examined in Section 5.3, we further conduct an ablation study on the remaining 6 datasets, as reported in Table 22. The notations used follow the same definitions as those in Table 4. In this extended analysis, we again observe that FracAug

consistently outperforms all four of its variants across the newly included datasets. This repeated pattern reinforces the importance of the individual components that constitute our framework and provides additional evidence that each design choice contributes meaningfully to the overall performance improvements achieved by FracAug.

## L MVP for Validation Set Only

Although we follow the standard data augmentation setting—using both the validation and test sets for pseudo-labeling—as adopted in related works such as ConsisGAD [2], we additionally evaluate a more restrictive setting in which only the validation set is used for pseudo-labeling. This complementary experiment allows us

**Table 16: Average AUROC, AUPRC, and F1-score on 6 datasets with multiple runs, using graph classification models as baselines, where the white columns represent vanilla models and the "+FA" represent models augmented by FracAug.**

| Datasets | Metrics  | GCN +FA              | SAGE +FA             | GAT +FA              | GIN +FA              | LRGNN +FA            | GRDL +FA             | BiView +FA           |
|----------|----------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| MCF-7    | AUROC    | 0.5753 <b>0.5840</b> | 0.5801 <b>0.5930</b> | 0.5885 <b>0.6058</b> | 0.5867 <b>0.5976</b> | 0.5467 <b>0.6117</b> | 0.5867 <b>0.6197</b> | 0.5602 <b>0.6011</b> |
|          | AUPRC    | 0.3035 <b>0.3073</b> | 0.3316 <b>0.3340</b> | 0.3678 <b>0.3814</b> | 0.2830 <b>0.2971</b> | 0.2951 <b>0.3309</b> | 0.3038 <b>0.3469</b> | 0.2537 <b>0.2864</b> |
|          | F1-score | 0.4982 <b>0.5074</b> | 0.4798 <b>0.4963</b> | 0.4573 <b>0.4687</b> | 0.5366 <b>0.5421</b> | 0.4660 <b>0.5290</b> | 0.5147 <b>0.5254</b> | 0.5731 <b>0.5785</b> |
| MOLT-4   | AUROC    | 0.5531 <b>0.5650</b> | 0.5326 <b>0.5727</b> | 0.5403 <b>0.5721</b> | 0.5733 <b>0.5854</b> | 0.5495 <b>0.5851</b> | 0.5858 <b>0.5924</b> | 0.5625 <b>0.5791</b> |
|          | AUPRC    | 0.3183 <b>0.3217</b> | 0.1907 <b>0.2602</b> | 0.3456 <b>0.3667</b> | 0.2830 <b>0.3001</b> | 0.3047 <b>0.3175</b> | 0.3018 <b>0.3101</b> | 0.2378 <b>0.2504</b> |
|          | F1-score | 0.4501 <b>0.4626</b> | 0.5166 <b>0.5277</b> | 0.4096 <b>0.4313</b> | 0.5072 <b>0.5103</b> | 0.4570 <b>0.4939</b> | 0.5091 <b>0.5117</b> | 0.5707 <b>0.5720</b> |
| PC-3     | AUROC    | 0.5697 <b>0.5863</b> | 0.5986 <b>0.6119</b> | 0.5707 <b>0.5865</b> | 0.5969 <b>0.6119</b> | 0.5690 <b>0.6102</b> | 0.6044 <b>0.6202</b> | 0.5664 <b>0.6157</b> |
|          | AUPRC    | 0.2740 <b>0.2837</b> | 0.3154 <b>0.3248</b> | 0.3562 <b>0.3626</b> | 0.2797 <b>0.2893</b> | 0.2320 <b>0.3038</b> | 0.3381 <b>0.3459</b> | 0.2386 <b>0.2685</b> |
|          | F1-score | 0.4745 <b>0.4876</b> | 0.4751 <b>0.4841</b> | 0.4036 <b>0.4166</b> | 0.5063 <b>0.5205</b> | <b>0.5103</b> 0.5024 | 0.4615 <b>0.4750</b> | <b>0.5811</b> 0.5791 |
| SW-620   | AUROC    | 0.5662 <b>0.5839</b> | 0.5800 <b>0.5968</b> | 0.5633 <b>0.5870</b> | 0.5938 <b>0.6004</b> | 0.5758 <b>0.5946</b> | 0.6005 <b>0.6046</b> | 0.5377 <b>0.6037</b> |
|          | AUPRC    | 0.3134 <b>0.3229</b> | <b>0.3401</b> 0.3260 | 0.2481 <b>0.2587</b> | 0.2776 <b>0.2813</b> | <b>0.3506</b> 0.3386 | <b>0.2908</b> 0.2901 | 0.2012 <b>0.2564</b> |
|          | F1-score | 0.4406 <b>0.4541</b> | 0.4339 <b>0.4678</b> | 0.4923 <b>0.5187</b> | 0.5090 <b>0.5155</b> | 0.4190 <b>0.4536</b> | 0.5059 <b>0.5135</b> | 0.5496 <b>0.5643</b> |
| NCI-H23  | AUROC    | 0.5811 <b>0.5864</b> | 0.5765 <b>0.6105</b> | 0.5777 <b>0.6084</b> | 0.5897 <b>0.5968</b> | 0.6002 <b>0.6315</b> | 0.6161 <b>0.6271</b> | 0.5600 <b>0.5655</b> |
|          | AUPRC    | 0.2777 <b>0.2777</b> | 0.3197 <b>0.3207</b> | <b>0.2966</b> 0.2945 | 0.2566 <b>0.2659</b> | 0.2830 <b>0.3205</b> | 0.3034 <b>0.3069</b> | 0.2125 <b>0.2187</b> |
|          | F1-score | 0.4751 <b>0.4819</b> | 0.4333 <b>0.4740</b> | 0.4548 <b>0.4961</b> | 0.5059 <b>0.5073</b> | 0.4987 <b>0.5055</b> | 0.4983 <b>0.5112</b> | 0.5723 <b>0.5768</b> |
| OVCAR-8  | AUROC    | 0.5692 <b>0.5809</b> | 0.5763 <b>0.5836</b> | 0.5396 <b>0.5784</b> | 0.5935 <b>0.5963</b> | 0.5628 <b>0.5984</b> | 0.6230 <b>0.6311</b> | 0.5417 <b>0.5420</b> |
|          | AUPRC    | 0.3240 <b>0.3263</b> | 0.3391 <b>0.3423</b> | 0.2010 <b>0.2401</b> | 0.2573 <b>0.2612</b> | 0.3106 <b>0.3290</b> | 0.3042 <b>0.3129</b> | 0.2075 <b>0.2174</b> |
|          | F1-score | 0.4216 <b>0.4334</b> | 0.4163 <b>0.4221</b> | 0.4855 <b>0.5060</b> | 0.5118 <b>0.5123</b> | 0.4260 <b>0.4518</b> | 0.5087 <b>0.5119</b> | 0.5563 <b>0.5575</b> |

**Table 17: Average AUROC, AUPRC, and F1-score on 6 datasets with multiple runs, using GAD models as baselines, where the white columns represent vanilla models and the "+FA" represent models augmented by FracAug.**

| Datasets | Metrics  | iGAD +FA             | GmapAD +FA           | RQGNN +FA            | UniGAD +FA           | GADTrans +FA         |
|----------|----------|----------------------|----------------------|----------------------|----------------------|----------------------|
| MCF-7    | AUROC    | 0.5670 <b>0.5756</b> | 0.5159 <b>0.5342</b> | 0.5522 <b>0.5709</b> | 0.5380 <b>0.5480</b> | 0.5712 <b>0.5856</b> |
|          | AUPRC    | 0.3402 <b>0.3525</b> | 0.3521 <b>0.3577</b> | 0.2379 <b>0.2648</b> | 0.2405 <b>0.2527</b> | 0.3029 <b>0.3288</b> |
|          | F1-score | 0.4546 <b>0.4549</b> | 0.3776 <b>0.3961</b> | 0.5609 <b>0.5804</b> | 0.4959 <b>0.5008</b> | <b>0.4926</b> 0.4901 |
| MOLT-4   | AUROC    | 0.5562 <b>0.5573</b> | 0.5100 <b>0.5374</b> | 0.5576 <b>0.5696</b> | 0.5353 <b>0.5445</b> | 0.5605 <b>0.5635</b> |
|          | AUPRC    | 0.3088 <b>0.3092</b> | 0.2704 <b>0.2827</b> | 0.2302 <b>0.2468</b> | 0.2111 <b>0.2241</b> | 0.3194 <b>0.3221</b> |
|          | F1-score | 0.4621 <b>0.4636</b> | 0.4354 <b>0.4592</b> | 0.5644 <b>0.5718</b> | 0.5079 <b>0.5125</b> | 0.4586 <b>0.4602</b> |
| PC-3     | AUROC    | 0.5526 <b>0.5674</b> | 0.5112 <b>0.5266</b> | 0.5618 <b>0.6043</b> | 0.5496 <b>0.5559</b> | 0.5743 <b>0.6007</b> |
|          | AUPRC    | 0.1887 <b>0.2159</b> | 0.2999 <b>0.3094</b> | 0.2370 <b>0.2663</b> | 0.3875 <b>0.3889</b> | <b>0.3999</b> 0.3968 |
|          | F1-score | 0.5217 <b>0.5229</b> | 0.3845 <b>0.3938</b> | 0.5721 <b>0.5972</b> | 0.3461 <b>0.3542</b> | 0.3646 <b>0.4012</b> |
| SW-620   | AUROC    | 0.5641 <b>0.5776</b> | 0.5279 <b>0.5362</b> | 0.5428 <b>0.5692</b> | 0.5427 <b>0.5688</b> | 0.5658 <b>0.5806</b> |
|          | AUPRC    | 0.3332 <b>0.3701</b> | <b>0.3506</b> 0.3479 | 0.1936 <b>0.2219</b> | 0.2527 <b>0.2585</b> | 0.3503 <b>0.3545</b> |
|          | F1-score | <b>0.4207</b> 0.4029 | 0.3605 <b>0.3734</b> | 0.5530 <b>0.5654</b> | 0.4600 <b>0.4903</b> | 0.4072 <b>0.4214</b> |
| NCI-H23  | AUROC    | 0.5689 <b>0.5721</b> | 0.5289 <b>0.5489</b> | 0.5704 <b>0.6061</b> | 0.5694 <b>0.5860</b> | 0.5821 <b>0.5888</b> |
|          | AUPRC    | 0.2267 <b>0.2288</b> | 0.3274 <b>0.3401</b> | 0.2166 <b>0.2500</b> | 0.2733 <b>0.2756</b> | 0.3368 <b>0.3375</b> |
|          | F1-score | 0.5039 <b>0.5066</b> | 0.3710 <b>0.3827</b> | 0.5770 <b>0.5820</b> | 0.4645 <b>0.4831</b> | 0.4247 <b>0.4321</b> |
| OVCAR-8  | AUROC    | 0.5609 <b>0.5685</b> | 0.5209 <b>0.5243</b> | 0.5549 <b>0.5773</b> | 0.5360 <b>0.5445</b> | 0.5813 <b>0.5824</b> |
|          | AUPRC    | 0.2319 <b>0.2325</b> | <b>0.2863</b> 0.2836 | 0.1933 <b>0.2216</b> | <b>0.3196</b> 0.3112 | 0.3231 <b>0.3242</b> |
|          | F1-score | 0.4880 <b>0.4991</b> | 0.3992 <b>0.4054</b> | 0.5618 <b>0.5794</b> | 0.3873 <b>0.4043</b> | 0.4365 <b>0.4369</b> |

**Table 18: Average AUROC, AUPRC, and F1-score on 6 datasets with multiple runs, using graph-level augmentation models as baselines, where the white columns represent vanilla models and their own augmentation method, while the "+FA" represent vanilla models augmented by FracAug.**

| Datasets | Metrics  | MAAv   | NS            | SM     | +FA           | GLAv   | GLA    | +FA           | GMv    | GM     | +FA           | FGWMv         | FGWM   | +FA           | DPv    | DP     | +FA           |
|----------|----------|--------|---------------|--------|---------------|--------|--------|---------------|--------|--------|---------------|---------------|--------|---------------|--------|--------|---------------|
| MCF-7    | AUROC    | 0.5496 | <b>0.5727</b> | 0.5428 | 0.5695        | 0.5797 | 0.5735 | <b>0.6068</b> | 0.5730 | 0.5581 | <b>0.5935</b> | 0.5731        | 0.5502 | <b>0.5828</b> | 0.5360 | 0.5045 | <b>0.5897</b> |
|          | AUPRC    | 0.2346 | 0.2445        | 0.2224 | <b>0.2455</b> | 0.2558 | 0.2456 | <b>0.2944</b> | 0.2767 | 0.2864 | <b>0.3081</b> | 0.2720        | 0.2130 | <b>0.2945</b> | 0.1837 | 0.1279 | <b>0.2861</b> |
|          | F1-score | 0.5179 | 0.5635        | 0.5512 | <b>0.5721</b> | 0.5607 | 0.5609 | <b>0.5793</b> | 0.5186 | 0.4879 | <b>0.5220</b> | 0.5585        | 0.5283 | <b>0.5971</b> | 0.5369 | 0.4923 | <b>0.5391</b> |
| MOLT-4   | AUROC    | 0.5506 | 0.5391        | 0.5113 | <b>0.5663</b> | 0.5585 | 0.5578 | <b>0.5792</b> | 0.5637 | 0.5547 | <b>0.5771</b> | 0.5477        | 0.5308 | <b>0.6067</b> | 0.5640 | 0.5755 | <b>0.5940</b> |
|          | AUPRC    | 0.2203 | 0.1914        | 0.1796 | <b>0.2356</b> | 0.2177 | 0.2159 | <b>0.2511</b> | 0.2411 | 0.2176 | <b>0.2559</b> | 0.1994        | 0.1939 | <b>0.3104</b> | 0.2446 | 0.2557 | <b>0.2896</b> |
|          | F1-score | 0.5595 | 0.5392        | 0.5048 | <b>0.5658</b> | 0.5540 | 0.5519 | <b>0.5733</b> | 0.5306 | 0.5368 | <b>0.5416</b> | <b>0.5446</b> | 0.5115 | <b>0.5366</b> | 0.5275 | 0.5281 | <b>0.5361</b> |
| PC-3     | AUROC    | 0.5688 | 0.5805        | 0.5284 | <b>0.5821</b> | 0.6207 | 0.5938 | <b>0.6221</b> | 0.5705 | 0.5646 | <b>0.5782</b> | 0.5490        | 0.5442 | <b>0.6107</b> | 0.5592 | 0.5710 | <b>0.6011</b> |
|          | AUPRC    | 0.2112 | 0.2233        | 0.1854 | <b>0.2385</b> | 0.2770 | 0.2386 | <b>0.2790</b> | 0.3506 | 0.3456 | <b>0.3587</b> | 0.2028        | 0.2108 | <b>0.2623</b> | 0.2542 | 0.2531 | <b>0.2765</b> |
|          | F1-score | 0.5698 | 0.5685        | 0.5321 | <b>0.5848</b> | 0.5650 | 0.5569 | <b>0.5686</b> | 0.4085 | 0.4058 | <b>0.4101</b> | 0.5027        | 0.4914 | <b>0.5633</b> | 0.4775 | 0.4852 | <b>0.5165</b> |
| SW-620   | AUROC    | 0.5577 | 0.5776        | 0.5232 | <b>0.5834</b> | 0.5822 | 0.5799 | <b>0.5936</b> | 0.5839 | 0.5722 | <b>0.5987</b> | 0.5265        | 0.5395 | <b>0.6183</b> | 0.5802 | 0.5903 | <b>0.6127</b> |
|          | AUPRC    | 0.2026 | 0.2205        | 0.1958 | <b>0.2279</b> | 0.2315 | 0.2258 | <b>0.2455</b> | 0.2599 | 0.2511 | <b>0.2728</b> | 0.1340        | 0.1777 | <b>0.2759</b> | 0.2578 | 0.2776 | <b>0.3035</b> |
|          | F1-score | 0.5641 | 0.5477        | 0.5273 | <b>0.5676</b> | 0.5422 | 0.5473 | <b>0.5786</b> | 0.5111 | 0.5015 | <b>0.5220</b> | 0.5283        | 0.5113 | <b>0.5729</b> | 0.5072 | 0.5002 | <b>0.5119</b> |
| NCI-H23  | AUROC    | 0.5792 | 0.5634        | 0.5323 | <b>0.5979</b> | 0.5773 | 0.5762 | <b>0.6194</b> | 0.5912 | 0.5647 | <b>0.6014</b> | 0.5658        | 0.5760 | <b>0.6422</b> | 0.5851 | 0.5804 | <b>0.5929</b> |
|          | AUPRC    | 0.2273 | 0.1983        | 0.2017 | <b>0.2407</b> | 0.2082 | 0.2188 | <b>0.2680</b> | 0.2587 | 0.2139 | <b>0.2672</b> | 0.1986        | 0.2345 | <b>0.3105</b> | 0.2340 | 0.2333 | <b>0.2355</b> |
|          | F1-score | 0.5821 | 0.5625        | 0.5445 | <b>0.5835</b> | 0.5659 | 0.5779 | <b>0.5914</b> | 0.5058 | 0.5087 | <b>0.5134</b> | <b>0.5561</b> | 0.5066 | <b>0.5358</b> | 0.5222 | 0.5225 | <b>0.5366</b> |
| OVCAR-8  | AUROC    | 0.5507 | 0.5494        | 0.5278 | <b>0.5726</b> | 0.5911 | 0.5859 | <b>0.6049</b> | 0.5786 | 0.5725 | <b>0.6024</b> | 0.5696        | 0.5713 | <b>0.6317</b> | 0.5808 | 0.5851 | <b>0.5943</b> |
|          | AUPRC    | 0.1775 | 0.1633        | 0.1665 | <b>0.2132</b> | 0.2346 | 0.2196 | <b>0.2447</b> | 0.2764 | 0.2994 | <b>0.3072</b> | 0.2696        | 0.2401 | <b>0.3060</b> | 0.2393 | 0.2405 | <b>0.2449</b> |
|          | F1-score | 0.5461 | 0.5408        | 0.5337 | <b>0.5749</b> | 0.5350 | 0.5548 | <b>0.5728</b> | 0.4733 | 0.4469 | <b>0.4766</b> | 0.4831        | 0.4950 | <b>0.5211</b> | 0.5099 | 0.5108 | <b>0.5277</b> |

**Table 19: Standard Deviation of AUROC, AUPRC, and F1-score for Table 16.**

| Datasets | Metrics  | GCN +FA       | SAGE +FA      | GAT +FA       | GIN +FA       | LRGNN +FA     | GRDL +FA      |
|----------|----------|---------------|---------------|---------------|---------------|---------------|---------------|
| MCF-7    | AUROC    | 0.0027 0.0016 | 0.0011 0.0123 | 0.0144 0.0011 | 0.0015 0.0010 | 0.0011 0.0069 | 0.0128 0.0050 |
|          | AUPRC    | 0.0110 0.0016 | 0.0024 0.0050 | 0.0436 0.0465 | 0.0036 0.0030 | 0.0245 0.0135 | 0.0256 0.0007 |
|          | F1-score | 0.0136 0.0038 | 0.0006 0.0134 | 0.0215 0.0442 | 0.0012 0.0017 | 0.0182 0.0030 | 0.0041 0.0081 |
| MOLT-4   | AUROC    | 0.0080 0.0015 | 0.0233 0.0035 | 0.0028 0.0101 | 0.0043 0.0006 | 0.0122 0.0055 | 0.0076 0.0035 |
|          | AUPRC    | 0.0093 0.0032 | 0.0391 0.0059 | 0.0217 0.0024 | 0.0126 0.0052 | 0.0018 0.0107 | 0.0076 0.0037 |
|          | F1-score | 0.0023 0.0047 | 0.0116 0.0003 | 0.0233 0.0155 | 0.0047 0.0058 | 0.0171 0.0177 | 0.0045 0.0092 |
| PC-3     | AUROC    | 0.0132 0.0047 | 0.0093 0.0018 | 0.0284 0.0150 | 0.0015 0.0042 | 0.0117 0.0015 | 0.0163 0.0045 |
|          | AUPRC    | 0.0067 0.0027 | 0.0092 0.0043 | 0.0235 0.0269 | 0.0017 0.0049 | 0.0465 0.0093 | 0.0332 0.0004 |
|          | F1-score | 0.0109 0.0038 | 0.0037 0.0016 | 0.0124 0.0071 | 0.0040 0.0017 | 0.0258 0.0071 | 0.0098 0.0064 |
| SW-620   | AUROC    | 0.0063 0.0010 | 0.0008 0.0014 | 0.0216 0.0112 | 0.0038 0.0026 | 0.0054 0.0153 | 0.0087 0.0077 |
|          | AUPRC    | 0.0190 0.0035 | 0.0078 0.0045 | 0.0115 0.0040 | 0.0027 0.0001 | 0.0068 0.0106 | 0.0064 0.0004 |
|          | F1-score | 0.0088 0.0045 | 0.0082 0.0022 | 0.0196 0.0237 | 0.0086 0.0045 | 0.0130 0.0292 | 0.0068 0.0129 |
| NCI-H23  | AUROC    | 0.0030 0.0033 | 0.0013 0.0016 | 0.0062 0.0020 | 0.0016 0.0045 | 0.0306 0.0085 | 0.0075 0.0087 |
|          | AUPRC    | 0.0074 0.0015 | 0.0031 0.0006 | 0.0098 0.0074 | 0.0014 0.0071 | 0.0038 0.0197 | 0.0061 0.0136 |
|          | F1-score | 0.0026 0.0029 | 0.0043 0.0028 | 0.0009 0.0100 | 0.0037 0.0004 | 0.0411 0.0349 | 0.0048 0.0011 |
| OVCAR-8  | AUROC    | 0.0047 0.0008 | 0.0026 0.0010 | 0.0201 0.0061 | 0.0018 0.0014 | 0.0130 0.0008 | 0.0013 0.0009 |
|          | AUPRC    | 0.0017 0.0009 | 0.0037 0.0045 | 0.0220 0.0062 | 0.0033 0.0016 | 0.0061 0.0122 | 0.0052 0.0018 |
|          | F1-score | 0.0070 0.0002 | 0.0003 0.0030 | 0.0086 0.0153 | 0.0005 0.0006 | 0.0203 0.0101 | 0.0037 0.0036 |

**Table 20: Standard Deviation of AUROC, AUPRC, and F1-score for Table 17.**

| Datasets | Metrics  | iGAD +FA      | GmapAD +FA    | RQGNN +FA     | UniGAD +FA    |
|----------|----------|---------------|---------------|---------------|---------------|
| MCF-7    | AUROC    | 0.0020 0.0022 | 0.0006 0.0035 | 0.0167 0.0136 | 0.0069 0.0055 |
|          | AUPRC    | 0.0118 0.0016 | 0.0135 0.0086 | 0.0047 0.0042 | 0.0100 0.0025 |
|          | F1-score | 0.0078 0.0015 | 0.0138 0.0037 | 0.0136 0.0067 | 0.0165 0.0097 |
| MOLT-4   | AUROC    | 0.0033 0.0020 | 0.0021 0.0038 | 0.0041 0.0132 | 0.0011 0.0001 |
|          | AUPRC    | 0.0075 0.0009 | 0.0168 0.0082 | 0.0247 0.0047 | 0.0226 0.0009 |
|          | F1-score | 0.0105 0.0018 | 0.0100 0.0016 | 0.0122 0.0001 | 0.0139 0.0009 |
| PC-3     | AUROC    | 0.0246 0.0037 | 0.0096 0.0021 | 0.0264 0.0017 | 0.0040 0.0009 |
|          | AUPRC    | 0.0459 0.0074 | 0.0069 0.0049 | 0.0017 0.0018 | 0.0011 0.0005 |
|          | F1-score | 0.0006 0.0011 | 0.0172 0.0068 | 0.0182 0.0028 | 0.0040 0.0015 |
| SW-620   | AUROC    | 0.0033 0.0008 | 0.0020 0.0033 | 0.0088 0.0266 | 0.0002 0.0059 |
|          | AUPRC    | 0.0023 0.0023 | 0.0100 0.0025 | 0.0010 0.0172 | 0.0052 0.0036 |
|          | F1-score | 0.0020 0.0033 | 0.0075 0.0064 | 0.0075 0.0033 | 0.0011 0.0049 |
| NCI-H23  | AUROC    | 0.0081 0.0101 | 0.0102 0.0040 | 0.0068 0.0205 | 0.0015 0.0034 |
|          | AUPRC    | 0.0413 0.0409 | 0.0221 0.0075 | 0.0064 0.0223 | 0.0022 0.0018 |
|          | F1-score | 0.0260 0.0235 | 0.0089 0.0023 | 0.0038 0.0010 | 0.0001 0.0028 |
| OVCAR-8  | AUROC    | 0.0117 0.0044 | 0.0016 0.0051 | 0.0110 0.0013 | 0.0017 0.0093 |
|          | AUPRC    | 0.0375 0.0405 | 0.0028 0.0016 | 0.0007 0.0042 | 0.0035 0.0153 |
|          | F1-score | 0.0155 0.0301 | 0.0006 0.0070 | 0.0037 0.0024 | 0.0012 0.0028 |

to examine the robustness of our framework under an extremely limited auxiliary data scenario.

As shown in Table 23, pseudo-labeling only the validation set leads to a reduction in performance gains compared to our original setting. This behavior indicates that having access to more data for pseudo-labeling can more fully reveal the strength of our framework. At the same time, the results also demonstrate that even when supplied with very limited auxiliary data, our method can still extract meaningful supervisory signals and deliver performance improvements. This resilience under constrained conditions highlights the superiority of our framework, further confirming that FracAug remains effective even in extreme data augmentation settings.

## M Original Loss for Baselines

We conduct an additional set of experiments in which baseline models were evaluated under the original, unweighted loss functions,

without applying any rebalancing techniques or class frequency adjustments. The results are summarized in Table 24. As shown, most baseline models experience a substantial decline in performance when trained with their original loss in the imbalanced setting. This degradation is consistent across metrics and datasets, confirming that the performance of these models is sensitive to skewed class distributions. Importantly, this finding validates our experimental choice in the main paper, i.e., the weighted loss was introduced precisely to prevent the baselines from collapsing under severe imbalance, thereby ensuring a fair and meaningful comparison with our proposed method. Without such weighting, the baselines fail to capture minority class patterns effectively, leading to inflated majority class predictions and weakened anomaly or minority detection capability. These newly added results therefore reinforce the significance of using balanced training strategies and further highlight the robustness advantage of our approach.

**Table 21: Standard Deviation of AUROC, AUPRC, and F1-score for Table 18.**

| Datasets | Metrics  | MAAv   | NS     | SM     | +FA    | GLAv   | GLA    | +FA    | GMv    | GM     | +FA    | FGWMv  | FGWM   | +FA    |
|----------|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| MCF-7    | AUROC    | 0.0069 | 0.0062 | 0.0154 | 0.0003 | 0.0069 | 0.0208 | 0.0091 | 0.0216 | 0.0173 | 0.0015 | 0.0091 | 0.0427 | 0.0014 |
|          | AUPRC    | 0.0070 | 0.0094 | 0.0107 | 0.0016 | 0.0112 | 0.0320 | 0.0126 | 0.0226 | 0.0272 | 0.0018 | 0.0004 | 0.0854 | 0.0152 |
|          | F1-score | 0.0169 | 0.0018 | 0.0168 | 0.0014 | 0.0003 | 0.0070 | 0.0082 | 0.0144 | 0.0016 | 0.0006 | 0.0330 | 0.0096 | 0.0055 |
| MOLT-4   | AUROC    | 0.0080 | 0.0225 | 0.0059 | 0.0065 | 0.0135 | 0.0029 | 0.0004 | 0.0038 | 0.0058 | 0.0001 | 0.0057 | 0.0083 | 0.0146 |
|          | AUPRC    | 0.0110 | 0.0284 | 0.0313 | 0.0016 | 0.0206 | 0.0047 | 0.0013 | 0.0010 | 0.0105 | 0.0003 | 0.0096 | 0.0409 | 0.0116 |
|          | F1-score | 0.0076 | 0.0174 | 0.0110 | 0.0046 | 0.0042 | 0.0030 | 0.0017 | 0.0065 | 0.0013 | 0.0001 | 0.0041 | 0.0136 | 0.0148 |
| PC-3     | AUROC    | 0.0062 | 0.0181 | 0.0296 | 0.0104 | 0.0046 | 0.0195 | 0.0053 | 0.0144 | 0.0179 | 0.0009 | 0.0246 | 0.0141 | 0.0011 |
|          | AUPRC    | 0.0078 | 0.0201 | 0.0329 | 0.0120 | 0.0091 | 0.0277 | 0.0099 | 0.0065 | 0.0290 | 0.0174 | 0.0651 | 0.0373 | 0.0017 |
|          | F1-score | 0.0033 | 0.0003 | 0.0430 | 0.0058 | 0.0111 | 0.0072 | 0.0153 | 0.0235 | 0.0053 | 0.0153 | 0.0163 | 0.0095 | 0.0001 |
| SW-620   | AUROC    | 0.0034 | 0.0043 | 0.0172 | 0.0020 | 0.0061 | 0.0135 | 0.0057 | 0.0029 | 0.0038 | 0.0008 | 0.0302 | 0.0241 | 0.0069 |
|          | AUPRC    | 0.0001 | 0.0059 | 0.0093 | 0.0031 | 0.0151 | 0.0141 | 0.0035 | 0.0191 | 0.0136 | 0.0002 | 0.0645 | 0.0704 | 0.0091 |
|          | F1-score | 0.0004 | 0.0027 | 0.0276 | 0.0019 | 0.0091 | 0.0182 | 0.0052 | 0.0142 | 0.0066 | 0.0013 | 0.0335 | 0.0169 | 0.0071 |
| NCI-H23  | AUROC    | 0.0139 | 0.0330 | 0.0020 | 0.0013 | 0.0057 | 0.0077 | 0.0061 | 0.0317 | 0.0018 | 0.0002 | 0.0551 | 0.0073 | 0.0045 |
|          | AUPRC    | 0.0125 | 0.0355 | 0.0049 | 0.0008 | 0.0118 | 0.0076 | 0.0085 | 0.0401 | 0.0070 | 0.0069 | 0.0661 | 0.0037 | 0.0071 |
|          | F1-score | 0.0059 | 0.0175 | 0.0029 | 0.0006 | 0.0100 | 0.0032 | 0.0037 | 0.0083 | 0.0036 | 0.0071 | 0.0305 | 0.0074 | 0.0205 |
| OVCAR-8  | AUROC    | 0.0031 | 0.0111 | 0.0193 | 0.0036 | 0.0023 | 0.0230 | 0.0007 | 0.0169 | 0.0099 | 0.0006 | 0.0194 | 0.0144 | 0.0063 |
|          | AUPRC    | 0.0171 | 0.0175 | 0.0139 | 0.0031 | 0.0083 | 0.0335 | 0.0008 | 0.0361 | 0.0016 | 0.0047 | 0.0783 | 0.0284 | 0.0034 |
|          | F1-score | 0.0221 | 0.0028 | 0.0255 | 0.0013 | 0.0072 | 0.0062 | 0.0004 | 0.0098 | 0.0131 | 0.0035 | 0.1100 | 0.0049 | 0.0072 |

**Table 22: Ablation study.**

| Datasets | Metrics  | GIN    | +FA    | w/o largest | w/o smallest | w/o WDML | w/o MVP |
|----------|----------|--------|--------|-------------|--------------|----------|---------|
| MCF-7    | AUROC    | 0.5867 | 0.5976 | 0.5848      | 0.5889       | 0.5860   | 0.5835  |
|          | AUPRC    | 0.2830 | 0.2971 | 0.2842      | 0.2882       | 0.2813   | 0.2790  |
|          | F1-score | 0.5366 | 0.5421 | 0.5317      | 0.5351       | 0.5372   | 0.5350  |
| MOLT-4   | AUROC    | 0.5733 | 0.5854 | 0.5770      | 0.5760       | 0.5772   | 0.5754  |
|          | AUPRC    | 0.2830 | 0.3001 | 0.2974      | 0.2862       | 0.2981   | 0.2881  |
|          | F1-score | 0.5072 | 0.5103 | 0.5001      | 0.5085       | 0.4998   | 0.5060  |
| PC-3     | AUROC    | 0.5969 | 0.6119 | 0.5963      | 0.6018       | 0.6026   | 0.5975  |
|          | AUPRC    | 0.2797 | 0.2893 | 0.2797      | 0.2815       | 0.2806   | 0.2780  |
|          | F1-score | 0.5063 | 0.5205 | 0.5055      | 0.5124       | 0.5145   | 0.5092  |
| SW-620   | AUROC    | 0.5938 | 0.6004 | 0.5941      | 0.5938       | 0.5949   | 0.5930  |
|          | AUPRC    | 0.2776 | 0.2813 | 0.2737      | 0.2778       | 0.2800   | 0.2697  |
|          | F1-score | 0.5090 | 0.5155 | 0.5132      | 0.5089       | 0.5082   | 0.5155  |
| NCI-H23  | AUROC    | 0.5897 | 0.5968 | 0.5893      | 0.5911       | 0.5866   | 0.5925  |
|          | AUPRC    | 0.2566 | 0.2659 | 0.2564      | 0.2633       | 0.2614   | 0.2656  |
|          | F1-score | 0.5059 | 0.5073 | 0.5054      | 0.5013       | 0.4968   | 0.5013  |
| OVCAR-8  | AUROC    | 0.5935 | 0.5963 | 0.5911      | 0.5918       | 0.5911   | 0.5904  |
|          | AUPRC    | 0.2573 | 0.2612 | 0.2579      | 0.2565       | 0.2573   | 0.2605  |
|          | F1-score | 0.5118 | 0.5123 | 0.5074      | 0.5100       | 0.5081   | 0.5038  |

**Table 23: Average AUROC, AUPRC, and F1-score on 3 datasets with multiple runs, where the "+FA" represents our original setting, and onlyval represents pseudo-labeling only the validation set.**

| Datasets      | Metrics  | GAT    | +FA    | onlyval | GIN    | +FA    | onlyval | iGAD   | +FA    | onlyval | MAAv   | +FA    | onlyval |
|---------------|----------|--------|--------|---------|--------|--------|---------|--------|--------|---------|--------|--------|---------|
| UACC257       | AUROC    | 0.5890 | 0.6174 | 0.6021  | 0.5877 | 0.6015 | 0.5946  | 0.5697 | 0.5748 | 0.5721  | 0.5623 | 0.5947 | 0.5731  |
|               | AUPRC    | 0.3493 | 0.3389 | 0.3250  | 0.2480 | 0.2598 | 0.2631  | 0.1906 | 0.1970 | 0.1956  | 0.1805 | 0.2210 | 0.1847  |
|               | F1-score | 0.4031 | 0.4455 | 0.4491  | 0.4906 | 0.4983 | 0.4908  | 0.5201 | 0.5224 | 0.5214  | 0.5541 | 0.5784 | 0.5582  |
| PROTEINS_full | AUROC    | 0.6157 | 0.6836 | 0.6633  | 0.5799 | 0.6174 | 0.6015  | 0.5976 | 0.6206 | 0.6125  | 0.6009 | 0.6217 | 0.6147  |
|               | AUPRC    | 0.6350 | 0.7005 | 0.6933  | 0.6259 | 0.6358 | 0.6261  | 0.6200 | 0.6333 | 0.6271  | 0.6183 | 0.6366 | 0.6357  |
|               | F1-score | 0.6158 | 0.6859 | 0.6508  | 0.5679 | 0.6175 | 0.6024  | 0.5960 | 0.6211 | 0.6137  | 0.6015 | 0.6214 | 0.614   |
| DBLP_v1       | AUROC    | 0.6119 | 0.6885 | 0.6489  | 0.6231 | 0.8044 | 0.7844  | 0.7755 | 0.7909 | 0.7901  | 0.6446 | 0.6822 | 0.6712  |
|               | AUPRC    | 0.7507 | 0.7796 | 0.7768  | 0.7201 | 0.8626 | 0.8446  | 0.8377 | 0.8473 | 0.8466  | 0.7689 | 0.7816 | 0.7814  |
|               | F1-score | 0.5782 | 0.6868 | 0.6311  | 0.5996 | 0.8028 | 0.7837  | 0.7749 | 0.7910 | 0.7902  | 0.6252 | 0.6778 | 0.6621  |

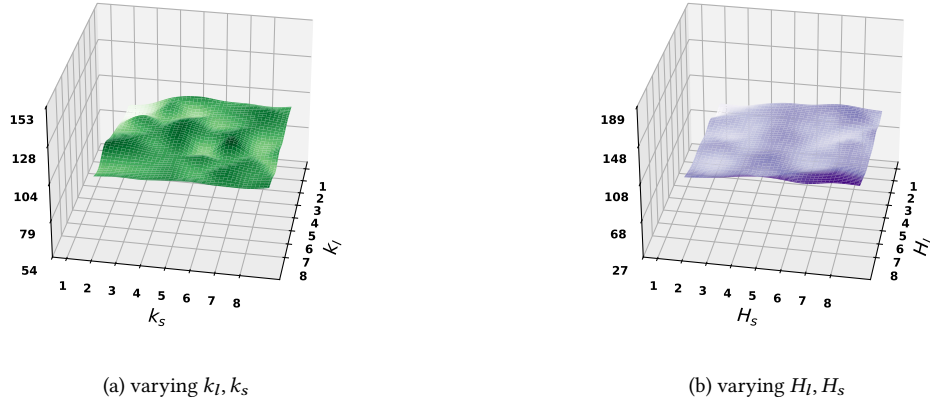
## N Running Time when Varying Hyperparameters

Because FGg involves four hyperparameters,  $k_l$ ,  $k_s$ ,  $H_l$ ,  $H_s$ , that might influence the running time, we evaluate the computational

overhead under different configurations. We present two 3D plots that report the runtime when varying hyperparameters, as shown in Figures 7. The first plot sweeps  $k_l$  and  $k_s$  from 1 to 8 and records the training time. The second plot similarly varies  $H_l$  and  $H_s$  from 1 to 8 and measures the runtime during training.

**Table 24: Average AUROC, AUPRC, and F1-score on 3 datasets with multiple runs, where the "+FA" represents our original setting, and oriloss represents the original setting of baselines.**

| Datasets      | Metrics  | GAT +FA | oriloss | GIN +FA | oriloss | MAA +FA | oriloss | GM +FA | oriloss |
|---------------|----------|---------|---------|---------|---------|---------|---------|--------|---------|
| UACC257       | AUROC    | 0.5890  | 0.6174  | 0.5583  | 0.5877  | 0.6015  | 0.5749  | 0.5023 | 0.5947  |
|               | AUPRC    | 0.3493  | 0.3389  | 0.3106  | 0.2480  | 0.2598  | 0.2448  | 0.0559 | 0.2210  |
|               | F1-score | 0.4031  | 0.4455  | 0.4039  | 0.4906  | 0.4983  | 0.4772  | 0.5005 | 0.5784  |
| PROTEINS_full | AUROC    | 0.6157  | 0.6836  | 0.5739  | 0.5799  | 0.6174  | 0.5517  | 0.6083 | 0.6217  |
|               | AUPRC    | 0.6350  | 0.7005  | 0.6097  | 0.6259  | 0.6358  | 0.5824  | 0.6294 | 0.6366  |
|               | F1-score | 0.6158  | 0.6859  | 0.5695  | 0.5679  | 0.6175  | 0.5503  | 0.6039 | 0.6214  |
| DBLP_v1       | AUROC    | 0.6119  | 0.6885  | 0.6263  | 0.6231  | 0.8044  | 0.6151  | 0.6608 | 0.6822  |
|               | AUPRC    | 0.7507  | 0.7796  | 0.7168  | 0.7201  | 0.8626  | 0.7154  | 0.7868 | 0.7816  |
|               | F1-score | 0.5782  | 0.6868  | 0.6199  | 0.5996  | 0.8028  | 0.6147  | 0.6408 | 0.6778  |

**Figure 7: Training time when varying  $k_l, k_s, H_l, H_s$  for UACC257 based on GIN.**

Across both experiments, we observe that the runtime does increase as the hyperparameters grow, but the increase is small and progresses gradually. Larger  $k_l$  and  $k_s$  require computing additional eigenvalues, and larger  $H_l$  and  $H_s$  apply more repeated or deeper fractional operations. Despite this, the overall time remains low and exhibits no sharp growth. These results confirm that FGG’s computational cost scales smoothly with its hyperparameters, and even at higher settings, it maintains an efficient and manageable runtime suitable for practical graph-level anomaly detection.

## O Replacement for FGG/MVP

To further evaluate the contribution of FracAug’s main components, we conduct an additional ablation study by replacing FGG. We substitute FGG with two standard perturbation-based augmentations, Node Drop and Edge Drop. As shown in Table 25, these replacements often result in degraded performance compared to our original setting, and sometimes even lower than their baselines. Although our MVP can rule out the noise from non-semantic-preserving generation by mutual verification to some extent, the fail to leverage guaranteed semantic-preserving properties prevents existing data augmentation methods from achieving optimal results, which further demonstrates the effectiveness of our proposed components.

We further replace MVP with a soft ensemble pseudo-labeling strategy that averages the predictions of the original and synthetic samples. This relaxation produces noisier and less reliable pseudo-labels, since it lacks the strict consistency constraint enforced by mutual verification. Consequently, as shown in Table 25, performance becomes unstable or deteriorates across datasets. In contrast, the full FracAug framework, using FGG for high-fidelity synthetic generation and MVP for robust pseudo-label filtering, consistently delivers superior results, underscoring the critical importance of both design components.

Instead of conducting experiments on GIN, we also conduct the same experiments on different backbones to prove that the effectiveness of our framework is a general advantage.

In conclusion, the additional ablation study, where FGG is replaced with either Node Drop or Edge Drop, and MVP is replaced with Soft Ensemble, further reinforces the effectiveness of each individual component. These controlled substitutions consistently lead to inferior performance, thereby validating that all the components contribute uniquely and substantially to the overall design.

## P Influence of Middle Eigenvalues

Originally, FGG is intentionally constructed around the largest top- $k_l$  and smallest top- $k_s$  eigenvalues, as these parts of the spectrum encode global structural patterns and localized irregularities, both

**Table 25: Ablation study by replacing FGG by Node Drop(ND) or Edge Drop (ED), and replacing MVP by Soft Ensemble (SE).**

| Datasets      | Metrics  | GIN    | +FA    | ND     | ED     | SE     | NS     | +FA    | ND     | ED     | SE     | GMixup | +FA    | ND     | ED     | SE     |
|---------------|----------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| MCF-7         | AUROC    | 0.5867 | 0.5976 | 0.5878 | 0.5838 | 0.5917 | 0.5496 | 0.5695 | 0.5575 | 0.5442 | 0.5594 | 0.5730 | 0.5935 | 0.5758 | 0.5720 | 0.5738 |
|               | AUPRC    | 0.2830 | 0.2971 | 0.2834 | 0.2832 | 0.2959 | 0.2346 | 0.2455 | 0.2224 | 0.2054 | 0.2248 | 0.2767 | 0.3081 | 0.2941 | 0.2947 | 0.2762 |
|               | F1-score | 0.5366 | 0.5421 | 0.5384 | 0.5309 | 0.5317 | 0.5179 | 0.5721 | 0.5460 | 0.5506 | 0.5589 | 0.5186 | 0.5220 | 0.5070 | 0.5009 | 0.5203 |
| MOLT-4        | AUROC    | 0.5733 | 0.5854 | 0.5790 | 0.5774 | 0.5789 | 0.5506 | 0.5663 | 0.5496 | 0.5521 | 0.5557 | 0.5637 | 0.5771 | 0.5737 | 0.5707 | 0.5714 |
|               | AUPRC    | 0.2830 | 0.3001 | 0.2967 | 0.2929 | 0.2949 | 0.2203 | 0.2356 | 0.2271 | 0.2261 | 0.2240 | 0.2411 | 0.2559 | 0.2414 | 0.2545 | 0.2459 |
|               | F1-score | 0.5072 | 0.5103 | 0.5024 | 0.5047 | 0.5101 | 0.5595 | 0.5658 | 0.5606 | 0.5622 | 0.5632 | 0.5306 | 0.5416 | 0.5378 | 0.5300 | 0.5414 |
| PC-3          | AUROC    | 0.5969 | 0.6119 | 0.6036 | 0.5991 | 0.6015 | 0.5688 | 0.5821 | 0.5664 | 0.5639 | 0.5679 | 0.5705 | 0.5782 | 0.5677 | 0.5700 | 0.5731 |
|               | AUPRC    | 0.2797 | 0.2893 | 0.2855 | 0.2837 | 0.2778 | 0.2112 | 0.2385 | 0.2132 | 0.2103 | 0.2149 | 0.3506 | 0.3587 | 0.3473 | 0.3527 | 0.3546 |
|               | F1-score | 0.5063 | 0.5205 | 0.5110 | 0.5058 | 0.5156 | 0.5698 | 0.5848 | 0.5716 | 0.5700 | 0.5725 | 0.4085 | 0.4101 | 0.4083 | 0.4038 | 0.4096 |
| SW-620        | AUROC    | 0.5938 | 0.6004 | 0.5919 | 0.5958 | 0.5932 | 0.5577 | 0.5834 | 0.5634 | 0.5610 | 0.5581 | 0.5839 | 0.5987 | 0.5855 | 0.5834 | 0.5872 |
|               | AUPRC    | 0.2776 | 0.2813 | 0.2721 | 0.2793 | 0.2746 | 0.2026 | 0.2279 | 0.2072 | 0.2041 | 0.1991 | 0.2599 | 0.2728 | 0.2574 | 0.2609 | 0.2658 |
|               | F1-score | 0.5090 | 0.5155 | 0.5112 | 0.5101 | 0.5109 | 0.5641 | 0.5676 | 0.5666 | 0.5651 | 0.5625 | 0.5111 | 0.5220 | 0.5169 | 0.5090 | 0.5110 |
| NCI-H23       | AUROC    | 0.5897 | 0.5968 | 0.5847 | 0.5850 | 0.5900 | 0.5792 | 0.5979 | 0.5729 | 0.5780 | 0.5909 | 0.5912 | 0.6014 | 0.5911 | 0.5947 | 0.5984 |
|               | AUPRC    | 0.2566 | 0.2659 | 0.2543 | 0.2522 | 0.2626 | 0.2273 | 0.2407 | 0.2255 | 0.2258 | 0.2295 | 0.2587 | 0.2672 | 0.2522 | 0.2665 | 0.2610 |
|               | F1-score | 0.5059 | 0.5073 | 0.5007 | 0.5031 | 0.5004 | 0.5821 | 0.5835 | 0.5816 | 0.5821 | 0.5776 | 0.5058 | 0.5134 | 0.5127 | 0.5036 | 0.5121 |
| OVCAR-8       | AUROC    | 0.5935 | 0.5963 | 0.5926 | 0.5918 | 0.5933 | 0.5507 | 0.5726 | 0.5603 | 0.5561 | 0.5549 | 0.5786 | 0.6024 | 0.5846 | 0.5840 | 0.5827 |
|               | AUPRC    | 0.2573 | 0.2612 | 0.2527 | 0.2584 | 0.2604 | 0.1775 | 0.2132 | 0.1904 | 0.1758 | 0.1740 | 0.2764 | 0.3072 | 0.2916 | 0.2851 | 0.2883 |
|               | F1-score | 0.5118 | 0.5123 | 0.5068 | 0.5081 | 0.5082 | 0.5461 | 0.5749 | 0.5628 | 0.5518 | 0.5359 | 0.4733 | 0.4766 | 0.4680 | 0.4729 | 0.4740 |
| P388          | AUROC    | 0.5565 | 0.5913 | 0.5662 | 0.5749 | 0.5645 | 0.5500 | 0.5720 | 0.5507 | 0.5700 | 0.5507 | 0.5469 | 0.5647 | 0.5467 | 0.5486 | 0.5484 |
|               | AUPRC    | 0.2850 | 0.3309 | 0.2933 | 0.3026 | 0.2948 | 0.1958 | 0.2229 | 0.1919 | 0.2077 | 0.2015 | 0.1694 | 0.1957 | 0.1821 | 0.1795 | 0.1789 |
|               | F1-score | 0.4468 | 0.4491 | 0.4445 | 0.4439 | 0.4480 | 0.5520 | 0.5746 | 0.5599 | 0.5678 | 0.5625 | 0.5315 | 0.5373 | 0.5313 | 0.5222 | 0.5224 |
| SF-295        | AUROC    | 0.5844 | 0.6076 | 0.5948 | 0.5972 | 0.5998 | 0.5649 | 0.5753 | 0.5636 | 0.5721 | 0.5624 | 0.5665 | 0.6040 | 0.5862 | 0.5841 | 0.5694 |
|               | AUPRC    | 0.2766 | 0.2832 | 0.2673 | 0.2796 | 0.2780 | 0.2114 | 0.2252 | 0.2154 | 0.2072 | 0.2149 | 0.2004 | 0.2509 | 0.2374 | 0.2278 | 0.2090 |
|               | F1-score | 0.4803 | 0.5047 | 0.5016 | 0.4932 | 0.4984 | 0.5736 | 0.5820 | 0.5752 | 0.5717 | 0.5745 | 0.5245 | 0.5371 | 0.5186 | 0.5260 | 0.5198 |
| SN12C         | AUROC    | 0.5995 | 0.6079 | 0.5972 | 0.5959 | 0.6017 | 0.5509 | 0.5795 | 0.5561 | 0.5538 | 0.5690 | 0.5713 | 0.5984 | 0.5882 | 0.5740 | 0.5817 |
|               | AUPRC    | 0.2696 | 0.2746 | 0.2677 | 0.2682 | 0.2695 | 0.1726 | 0.2164 | 0.1710 | 0.1704 | 0.1855 | 0.2163 | 0.2524 | 0.2386 | 0.2150 | 0.2328 |
|               | F1-score | 0.5030 | 0.5110 | 0.5017 | 0.4991 | 0.5065 | 0.5538 | 0.5770 | 0.5490 | 0.5523 | 0.5650 | 0.5136 | 0.5195 | 0.5175 | 0.5185 | 0.5127 |
| UACC257       | AUROC    | 0.5877 | 0.6015 | 0.5905 | 0.5859 | 0.5952 | 0.5623 | 0.5947 | 0.5768 | 0.5679 | 0.5689 | 0.5853 | 0.6209 | 0.5989 | 0.5945 | 0.5951 |
|               | AUPRC    | 0.2480 | 0.2598 | 0.2442 | 0.2553 | 0.2573 | 0.1805 | 0.2210 | 0.1932 | 0.1813 | 0.2112 | 0.2365 | 0.2963 | 0.2405 | 0.2548 | 0.2727 |
|               | F1-score | 0.4906 | 0.4983 | 0.4978 | 0.4818 | 0.4919 | 0.5541 | 0.5784 | 0.5657 | 0.5615 | 0.5794 | 0.4993 | 0.4898 | 0.4883 | 0.4933 | 0.4780 |
| PROTEINS_full | AUROC    | 0.5799 | 0.6174 | 0.5808 | 0.5815 | 0.5979 | 0.6009 | 0.6217 | 0.6054 | 0.6039 | 0.6157 | 0.5411 | 0.6097 | 0.5759 | 0.5523 | 0.5554 |
|               | AUPRC    | 0.6259 | 0.6358 | 0.6139 | 0.6075 | 0.6297 | 0.6183 | 0.6366 | 0.6271 | 0.6353 | 0.6304 | 0.5810 | 0.6244 | 0.5997 | 0.5770 | 0.5992 |
|               | F1-score | 0.5679 | 0.6175 | 0.5770 | 0.5801 | 0.5933 | 0.6015 | 0.6214 | 0.6003 | 0.5991 | 0.6137 | 0.5348 | 0.6102 | 0.5753 | 0.5520 | 0.5494 |
| DBLP_v1       | AUROC    | 0.6231 | 0.8044 | 0.7034 | 0.6791 | 0.7849 | 0.6446 | 0.6822 | 0.6601 | 0.6414 | 0.6712 | 0.7939 | 0.7994 | 0.7891 | 0.7803 | 0.7812 |
|               | AUPRC    | 0.7201 | 0.8626 | 0.8003 | 0.7738 | 0.8411 | 0.7689 | 0.7816 | 0.7794 | 0.7329 | 0.7614 | 0.8503 | 0.8563 | 0.8479 | 0.8394 | 0.8403 |
|               | F1-score | 0.5996 | 0.8028 | 0.7009 | 0.6775 | 0.7847 | 0.6252 | 0.6778 | 0.6467 | 0.6399 | 0.6712 | 0.7937 | 0.7985 | 0.7885 | 0.7805 | 0.7812 |

of which are crucial for effectively distinguishing anomalous graphs. To further examine this design choice, we conduct an analysis to determine whether incorporating the middle region of the eigenvalue spectrum offers any meaningful contribution to our fractional augmentation process.

As shown in Table 26, our results indicate that including all eigenvalues does not lead to any noticeable performance improvement across the evaluated datasets. Moreover, extracting these intermediate spectral components requires performing a full eigendecomposition, which incurs a computational complexity of  $O(n^3)$ , where  $n$  denotes the number of nodes in the graph. This stands in sharp contrast to the top- $k$  approximate eigensolvers used in our original design, whose complexity is only  $O(k \cdot nnz \cdot t)$ , with  $nnz$  representing the number of non-zero entries in the adjacency matrix and  $t$  denoting the iteration count of the eigendecomposition. Given the minimal performance gains and the substantially higher computational burden, these findings confirm that focusing solely on the largest top- $k_l$  and smallest top- $k_s$  eigenvalues is both an effective and efficient choice for FGG’s augmentation strategy.

## Q Comparison with warmup models

We conduct an additional experiment to further demonstrate the effectiveness of our proposed model. As described in Section 4, before integrating FracAug into the baseline models, we ensure that the baselines first acquire a foundational understanding of the graph-level anomaly detection dataset by warmup. This step is

crucial to stabilize their predictions and enhance the reliability of pseudo-labeling.

As shown in Table 27, the baseline models that undergo only a brief warmup phase, without sufficient training epochs, are outperformed by their fully trained counterparts. However, once FracAug is integrated into the training process of warmup baselines and then trained for an equivalent number of epochs as the fully-trained baselines, it becomes evident that the warmup models with FracAug achieve substantial performance improvements over the baselines. Ultimately, these models reach state-of-the-art performance, providing strong evidence of the effectiveness of our proposed FracAug framework.

## R Comparison with unsuervised and finetuned models

We first conduct comparisons between unsupervised graph-level anomaly detection models from a novel benchmark paper [15] and the FracAug-enhanced GIN. The results in Table 28 show that unsupervised training falls significantly behind our method, even though we only utilize 1% examples as the training set, indicating that label information plays a crucial role in achieving strong anomaly detection performance. Consistent with our earlier experiments in Appendix G, increasing the amount of available labeled data leads to further improvements, reinforcing the importance of label supervision. These findings highlight the importance and effectiveness of our proposed plug-in data augmentation framework, since

**Table 26: Ablation study for middle eigenvalues.**

| Datasets      | Metrics  | GIN    | +FA    | w/ middle |
|---------------|----------|--------|--------|-----------|
| MCF-7         | AUROC    | 0.5867 | 0.5976 | 0.5958    |
|               | AUPRC    | 0.2830 | 0.2971 | 0.2913    |
|               | F1-score | 0.5366 | 0.5421 | 0.5452    |
| MOLT-4        | AUROC    | 0.5733 | 0.5854 | 0.5851    |
|               | AUPRC    | 0.2830 | 0.3001 | 0.3034    |
|               | F1-score | 0.5072 | 0.5103 | 0.5066    |
| PC-3          | AUROC    | 0.5969 | 0.6119 | 0.6182    |
|               | AUPRC    | 0.2797 | 0.2893 | 0.2898    |
|               | F1-score | 0.5063 | 0.5205 | 0.5283    |
| SW-620        | AUROC    | 0.5938 | 0.6004 | 0.5980    |
|               | AUPRC    | 0.2776 | 0.2813 | 0.2820    |
|               | F1-score | 0.5090 | 0.5155 | 0.5110    |
| NCI-H23       | AUROC    | 0.5897 | 0.5968 | 0.5923    |
|               | AUPRC    | 0.2566 | 0.2659 | 0.2614    |
|               | F1-score | 0.5059 | 0.5073 | 0.5063    |
| OVCAR-8       | AUROC    | 0.5935 | 0.5963 | 0.6026    |
|               | AUPRC    | 0.2573 | 0.2612 | 0.2617    |
|               | F1-score | 0.5118 | 0.5123 | 0.5119    |
| P388          | AUROC    | 0.5565 | 0.5913 | 0.6031    |
|               | AUPRC    | 0.2850 | 0.3309 | 0.3376    |
|               | F1-score | 0.4468 | 0.4491 | 0.4483    |
| SF-295        | AUROC    | 0.5844 | 0.6076 | 0.6089    |
|               | AUPRC    | 0.2766 | 0.2832 | 0.2873    |
|               | F1-score | 0.4803 | 0.5047 | 0.4983    |
| SN12C         | AUROC    | 0.5995 | 0.6079 | 0.6082    |
|               | AUPRC    | 0.2696 | 0.2746 | 0.2756    |
|               | F1-score | 0.5030 | 0.5110 | 0.5103    |
| UACC257       | AUROC    | 0.5877 | 0.6015 | 0.5980    |
|               | AUPRC    | 0.2480 | 0.2598 | 0.2525    |
|               | F1-score | 0.4906 | 0.4983 | 0.5006    |
| PROTEINS_full | AUROC    | 0.5799 | 0.6174 | 0.6105    |
|               | AUPRC    | 0.6259 | 0.6358 | 0.6321    |
|               | F1-score | 0.5679 | 0.6175 | 0.6096    |
| DBLP_v1       | AUROC    | 0.6231 | 0.8044 | 0.8028    |
|               | AUPRC    | 0.7201 | 0.8626 | 0.8607    |
|               | F1-score | 0.5996 | 0.8028 | 0.8015    |

in real deployment, anomaly detection works usually suffer from the limited label supervision issue.

Furthermore, we compare our performance with Mole-BERT [18], which is a model pretrained in an unsupervised manner and subsequently finetuned with labeled data for a specific domain. As shown in Table 28, the unsupervised pretraining alone produces suboptimal results, consistent with the behavior of the unsupervised baselines. Even after finetuning with the same labeled data budget as our framework, the performance remains inferior to ours to a large extent. This observation suggests two key points: (1) finetuning may inherit a suboptimal parameter space induced by unsupervised pretraining, limiting its adaptability, and (2) sufficient and properly utilized label information is essential for achieving robust performance. Together, these findings further validate the novelty and effectiveness of our plug-in data augmentation framework, which can effectively generate more fidelity label information, even under extremely limited supervision.

## S Additional Hyperparameter Analysis

Beyond the results in Section 5.4, we also conduct hyperparameter analysis on DBLP\_v1, as shown in Figure 8. The experiment is conducted under the same setting as the one in Section 5.4. We can draw a similar conclusion that all metrics remain stable across this range, demonstrating that FracAug’s pseudo-labeling is robust to threshold selection.

## T Datasets from Other Domain

To further evaluate the generalizability of the proposed FracAug, we extend our experiments to two datasets from additional domains, where IMDB and REDDIT are both representative social network datasets. As reported in Table 29, FracAug consistently improves the performance of different baseline models on these datasets. These results indicate that the benefits of FracAug are not limited to a specific data domain or graph distribution. Instead, by generating semantic-preserving fractional graph variants and improving training under limited supervision, FracAug can serve as a broadly applicable plug-in augmentation framework across diverse graph-level anomaly detection scenarios.

## U Embedding-distribution Comparison

We report quantitative results that reflect the embedding-distribution comparison, as shown in 30. Specifically, let  $d_n$  ( $d_a$ ) denote the distance between the original and synthetic graphs in the normal (abnormal) class, and let  $d$  denote the distance between the synthetic normal and synthetic abnormal graphs. For a baseline model  $M$ , we define

$$M_n = \frac{d_n}{d}, \quad M_a = \frac{d_a}{d}.$$

A smaller value of  $M_n$  or  $M_a$  is preferred, because it means that the synthetic graphs stay close to the original graphs of the same class, while the two synthetic classes remain well separated. In other words, the within-class discrepancy is small relative to the between-class separation. Therefore, the small ratios reported above indicate that the generated graphs preserve the class-wise embedding distribution well and are consistent with the original graphs.

## V Scalability

To address the concern of scalability, we conducted new experiments on the public graph-level anomaly datasets with the largest per-graph sizes that we could identify. These new datasets therefore stress per-graph computational cost more directly than those in the main evaluation, as shown in Table 31. We compare FracAug with representative graph augmentation baselines, including NodeSam, SubMix, and FGWM. The time and memory are reported in the same way as in Section 5, where we provide the memory and time analysis: for FracAug-enhanced methods, the two terms denote the preprocessing and the FracAug cost, while single-term baselines report their total cost. The results show that FracAug remains effective while keeping competitive runtime and memory.

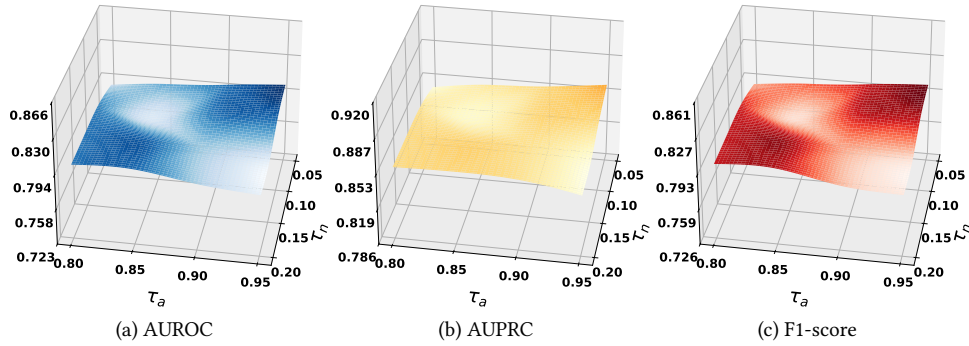
The results are presented in Table 32. On REDDIT-BINARY, each graph has on average 429.63 nodes and 497.75 edges, which is much larger per graph than the datasets in the main evaluation. FracAug substantially outperforms NodeSam, SubMix, and

**Table 27: Average AUROC, AUPRC, and F1-score on 3 datasets with multiple runs, where the "+FA" represents our original setting, and warmup represents the baseline mode after the warmup phase.**

| Datasets      | Metrics  | GAT +FA | warmup | GIN +FA | warmup | iGAD +FA | warmup | MAAv +FA | warmup |
|---------------|----------|---------|--------|---------|--------|----------|--------|----------|--------|
| UACC257       | AUROC    | 0.5890  | 0.6174 | 0.5568  | 0.5877 | 0.6015   | 0.5606 | 0.5697   | 0.5748 |
|               | AUPRC    | 0.3493  | 0.3389 | 0.3139  | 0.2480 | 0.2598   | 0.1994 | 0.1906   | 0.1970 |
|               | F1-score | 0.4031  | 0.4455 | 0.3993  | 0.4906 | 0.4983   | 0.4972 | 0.5201   | 0.5224 |
| PROTEINS_full | AUROC    | 0.6157  | 0.6836 | 0.6012  | 0.5799 | 0.6174   | 0.5723 | 0.5976   | 0.6206 |
|               | AUPRC    | 0.6350  | 0.7005 | 0.6112  | 0.6259 | 0.6358   | 0.5987 | 0.6200   | 0.6333 |
|               | F1-score | 0.6158  | 0.6859 | 0.6014  | 0.5679 | 0.6175   | 0.5713 | 0.5960   | 0.6211 |
| DBLP_v1       | AUROC    | 0.6119  | 0.6885 | 0.5436  | 0.6231 | 0.8044   | 0.6098 | 0.7755   | 0.7909 |
|               | AUPRC    | 0.7507  | 0.7796 | 0.7207  | 0.7201 | 0.8626   | 0.7014 | 0.8377   | 0.8473 |
|               | F1-score | 0.5782  | 0.6868 | 0.5425  | 0.5996 | 0.8028   | 0.5910 | 0.7749   | 0.7910 |

**Table 28: Comparison with unsupervised and finetuned models. "+FA" represents FracAug-enhanced GIN. "OOM" represents the out-of-memory issue.**

| Datasets      | Metrics  | Unsupervised |        |         |        |        |        | Mole-BERT |          | ours   |
|---------------|----------|--------------|--------|---------|--------|--------|--------|-----------|----------|--------|
|               |          | OCGIN        | GLADC  | GLocaKD | OCGTL  | SIGNET | CVTGAD | pretrain  | finetune |        |
| PROTEINS_full | AUROC    | 0.3435       | 0.3206 | 0.5596  | 0.5971 | 0.5841 | 0.2485 | 0.4414    | 0.5124   | 0.5799 |
|               | AUPRC    | 0.3132       | 0.3032 | 0.4941  | 0.4302 | 0.4962 | 0.2815 | 0.5968    | 0.5988   | 0.6259 |
|               | F1-score | 0.2880       | 0.2879 | 0.3733  | 0.2879 | 0.2879 | 0.2879 | 0.3697    | 0.4882   | 0.5679 |
| DBLP_v1       | AUROC    | 0.4536       | 0.5340 | 0.4370  | 0.5425 | 0.5131 | OOM    | 0.5077    | 0.6677   | 0.6231 |
|               | AUPRC    | 0.4651       | 0.5470 | 0.4522  | 0.5552 | 0.5332 | OOM    | 0.6506    | 0.7305   | 0.7201 |
|               | F1-score | 0.3290       | 0.3288 | 0.3288  | 0.3378 | 0.3378 | OOM    | 0.5069    | 0.6678   | 0.5996 |

**Figure 8: Varying  $\tau_a$  and  $\tau_n$  for DBLP\_v1 based on GIN.****Table 29: Performance of datasets from other domains.**

| Datasets | Metrics  | SAGE +FA | GIN +FA | RQGN +FA |
|----------|----------|----------|---------|----------|
| IMDB     | AUC      | 0.7007   | 0.7244  | 0.7070   |
|          | AUPRC    | 0.7760   | 0.7930  | 0.7797   |
|          | F1-score | 0.6974   | 0.7242  | 0.7069   |
| REDDIT   | AUC      | 0.7509   | 0.7665  | 0.6361   |
|          | AUPRC    | 0.8163   | 0.8265  | 0.7739   |
|          | F1-score | 0.7491   | 0.7651  | 0.6035   |

**Table 30: Embedding-distribution comparison results.**

| Datasets      | $GCN_n$ | $GCN_a$ | $SAGE_n$ | $SAGE_a$ | $GAT_n$ | $GAT_a$ |
|---------------|---------|---------|----------|----------|---------|---------|
| PROTEINS_full | 0.0640  | 0.0875  | 0.0334   | 0.1042   | 0.0286  | 0.0858  |
| DBLP_v1       | 0.0301  | 0.1034  | 0.0786   | 0.0509   | 0.0521  | 0.0504  |

FGWM in all metrics while maintaining practical runtime and memory. On github\_stargazers, whose graphs also have larger average node/edge counts than the main molecular benchmarks, FracAug remains competitive and improves over the corresponding vanilla augmentation baselines, while requiring less runtime than Node-Sam/SubMix and much less memory than several baselines.

Algorithmically, FracAug avoids full eigendecomposition: it uses Arnoldi-based top-k eigensolvers and retains only the top  $k_l$  largest and top  $k_s$  smallest eigenpairs, typically 3 or 4. The preprocessing is performed once and cached, so the heavier spectral step is amortized across training. These new experiments confirm that this design remains practical on the larger datasets, demonstrating the scalability of our method in real-world deployment.

Table 31: Statistics of scalability datasets.

| Dataset           | Normal | Abnormal | Abnormal Rate | Avg. Nodes | Avg. Edges | Feature Dim |
|-------------------|--------|----------|---------------|------------|------------|-------------|
| REDDIT-BINARY     | 1000   | 1000     | 0.5000        | 429.63     | 497.75     | 32          |
| github_stargazers | 6808   | 5917     | 0.4650        | 113.79     | 234.64     | 30          |

Table 32: Comparison of scalability datasets.

| Dataset           | Metric   | NSv+FA        | NodeSam | SubMix  | FGWMv+FA      | FGWM    |
|-------------------|----------|---------------|---------|---------|---------------|---------|
| REDDIT-BINARY     | AUROC    | 0.7423        | 0.5566  | 0.6148  | 0.6648        | 0.5750  |
|                   | AUPRC    | 0.8087        | 0.7416  | 0.7599  | 0.7779        | 0.6863  |
|                   | F1-score | 0.7412        | 0.4978  | 0.5828  | 0.6483        | 0.5747  |
|                   | Time     | 224.21+573.79 | 433.29  | 321.68  | 224.21+604.27 | 3876.28 |
|                   | Memory   | 118.73+360.66 | 678.58  | 917.90  | 118.73+350.59 | 644.71  |
| github_stargazers | AUROC    | 0.5230        | 0.5073  | 0.5074  | 0.5776        | 0.5381  |
|                   | AUPRC    | 0.6646        | 0.6433  | 0.6405  | 0.7570        | 0.6869  |
|                   | F1-score | 0.5216        | 0.5029  | 0.5070  | 0.5535        | 0.5381  |
|                   | Time     | 465.07+315.43 | 1606.48 | 1028.63 | 465.07+321.45 | 1171.99 |
|                   | Memory   | 161.67+542.59 | 700.80  | 787.42  | 161.67+303.37 | 720.18  |